Supplemental Material to
Thermal fluctuations set fundamental limits on ion channel function

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S1 Introduction

In this Supplemental Material, we show the explicit derivations for the expressions shown in the main text. Specifically, we compute the noise statistics for shot noise and Johnson-Nyquist noise on both a flat and a spherical membrane. We also characterize the communication channel in the spherical case.

S1.1 Notation

Throughout this document, we will use the following notation:

- Vectors $\mathbf{r}$ represent arbitrary coordinates in $\mathbb{R}^3$, while vectors $\mathbf{x}$ represent coordinates on the surface of the membrane (for both the flat and spherical membrane).
- The element $ds(\mathbf{x})$ represents the area element at the point $\mathbf{x}$ on the membrane. For example, if using spherical coordinates for the spherical membrane of radius R, then $ds(\mathbf{x}) = R^2 \sin(\theta)\, d\theta\, d\phi$.
- The operator ∂_n denotes the directional derivative in the direction of the normal to the membrane. We take the convention of the normal to be $\hat{\mathbf{z}}$ for the planar membrane and $\hat{\mathbf{r}}$ for the spherical one.
- Fourier transforms, in both space and time, are implicit. This means that $f(\omega)$ is understood to be the Fourier transform of $f(t)$. We adopt the following convention for the time transform:

$$f(\omega) = \int dt\, f(t) e^{-i\omega t}, \quad f(t) = \frac{1}{2\pi} \int d\omega\, f(\omega) e^{i\omega t},$$

  and similarly for the spatial transform. When transforming spatial coordinates, we explicitly state if only a subset of coordinates are being transformed.
- We use spherical harmonics for the analysis of the spherical membrane. We denote these with $Y_\ell^m(\mathbf{x})$, which corresponds to the value of the (ℓ, m) harmonic evaluated at the point $\mathbf{x}/R$ on the unit sphere. For a function $g(\mathbf{x})$, we define its harmonic components as

$$g_\ell^m = \frac{1}{R^2} \int ds(\mathbf{x})\, g(\mathbf{x}) Y_\ell^{m*}(\mathbf{x}).$$

S2 Charge dynamics in the Poisson-Nernst-Planck model

S2.1 Charge dynamics in the bulk

The electrical dynamics in the bulk are determined by the movement of charged ions. We use the Poisson-Nernst-Planck (PNP) model to describe the dynamics of these ions. We assume there are M ionic species, and that the state of the system is characterized by their concentrations $n_1(\mathbf{r}), \ldots, n_M(\mathbf{r})$. We begin by specifying the free energy functional

$$\mathcal{F}_0[\mathbf{n}] = U[\mathbf{n}] + \frac{1}{\beta} \sum_i \int d^3\mathbf{r}\, n_i(\mathbf{r}) \log(n_i(\mathbf{r})/n_0), \tag{S2.1}$$

where the first term is the potential energy from electrostatic interactions and the second term is an entropic contribution. The dynamics of this model are characterized by the transport of ions:

$$\partial_t n_i + \nabla \cdot \mathbf{j}_i = 0, \quad \mathbf{j}_i = -\beta D n_i \nabla \left(\frac{\delta \mathcal{F}_0}{\delta n_i} \right). \tag{S2.2}$$

Here D is the diffusion constant, which we assume to be the same for all ions for simplicity. These result in

$$\mathbf{j}_i = -D \nabla n_i + \beta D n_i z_i e \mathbf{E}. \tag{S2.3}$$

The net charge in the bulk is generated from deviations from the average bulk concentrations $\mathbf{n}^0$. These satisfy $\sum_i n_i^0 e z_i = 0$ because of neutrality, where z_i is the valence of species i. Specifically, we can write $n_i = n_i^0 + \hat{n}_i$. We can define the local density fluctuation and total current as

$$\rho(\mathbf{r}, t) \coloneqq \sum_i e z_i \hat{n}_i, \quad \mathbf{J} \coloneqq \sum_i e z_i \mathbf{j}_i. \tag{S2.4}$$

Note that he net charge fluctuations satisfy the continuity equation

$$\partial_t \rho + \nabla \cdot \mathbf{J} = 0. \tag{S2.5}$$

Since we consider small fluctuations, we have $\hat{n}_i \ll n_i^0$, and the total current is

$$\mathbf{J} = \alpha \mathbf{E} - D \nabla \rho, \tag{S2.6}$$

where the effective conductance is given by

$$\alpha \coloneqq \beta D \sum_i n_i^0 e^2 z_i^2. \tag{S2.7}$$

To close the dynamics of the charge density, we use the fact that the electric field satisfies Gauss's law

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon}, \tag{S2.8}$$

where ϵ is the permittivity of the bulk. The continuity equation, then, becomes

$$\tau_D \partial_t \rho = -\rho + l_D^2 \nabla^2 \rho, \tag{S2.9}$$

where the Debye time and the Debye length are, respectively, defined by

$$\tau_D \coloneqq \frac{\epsilon}{\alpha}, \quad l_D^2 \coloneqq \frac{D\epsilon}{\alpha}. \tag{S2.10}$$

In the cellular environment, we have $\tau_D \approx 10^{-9}$ s and $l_D \approx 7 \times 10^{-10}$ m.

To build intuition, we can analyze this equation in the absence of a membrane. Transforming this equation in space, we obtain

$$\partial_t \rho(\mathbf{k}) = -\frac{(1 + k^2 l_D^2)}{\tau_D} \rho(\mathbf{k}). \tag{S2.11}$$

This means that long wavelength modes (with $k l_D \ll 1$) decay uniformly with timescale τ_D, so charge gets quickly screened at distances greater than l_D.

S2.2 Electric potential in the presence of a membrane

We now analyze the effect of adding a membrane on the dynamics of the charge density. We model the membrane as a slab in the region $|z| < d/2$ with permittivity $\epsilon_m < \epsilon$.

First, we determine the potential generated by an arbitrary charge density profile. This will allow us to characterize the dynamics near the membrane. The potential solves the Poisson equation both inside and outside the membrane. Since there is no charge inside the membrane, this can be written as

$$\nabla^2 \Phi(\mathbf{r}) = \begin{cases} -\frac{\rho(\mathbf{r})}{\epsilon} & \text{if } |z| > d/2, \\ 0 & \text{if } |z| < d/2. \end{cases} \tag{S2.12}$$

To simplify the distinction between the regions, we denote the potential with Φ^+ in the region $z > d/2$, with Φ^- in the region $z < -d/2$ and with Φ^m in the region $|z| < d/2$. At the interfaces between the membrane

and the bulk, the normal component of the electric displacement field must be continuous. This yields the interface conditions (abusing notation, here $\mathbf{x}$ is a 2D vector representing coordinates on the membrane)

$$\begin{aligned}\epsilon\partial_n\Phi^+(\mathbf{x},d/2) &= \epsilon_m\partial_n\Phi^m(\mathbf{x},d/2) \coloneqq \lambda^+(\mathbf{x}),\\ \epsilon\partial_n\Phi^-(\mathbf{x},-d/2) &= \epsilon_m\partial_n\Phi^m(\mathbf{x},-d/2) \coloneqq \lambda^-(\mathbf{x}).\end{aligned} \tag{S2.13}$$

The fields $\lambda^\pm$ simplify the notation when characterizing the solution. Additionally, the potential must be continuous at the boundaries:

$$\Phi^+(\mathbf{x},d/2) = \Phi^m(\mathbf{x},d/2), \quad \Phi^-(\mathbf{x},-d/2) = \Phi^m(\mathbf{x},-d/2). \tag{S2.14}$$

To solve for the potential, we transform the coordinates parallel to the membrane. With this, the Poisson equation takes the form

$$(\partial_z^2 - k^2)\Phi(\mathbf{k},z) = \begin{cases} -\frac{\rho(\mathbf{k},z)}{\epsilon} & \text{if } |z| > d/2,\\ 0 & \text{if } |z| < d/2.\end{cases} \tag{S2.15}$$

Along with the boundary conditions, this fully determines the potential. For the bulk, we have

$$\begin{aligned}\Phi^+(\mathbf{k},z) &= -\frac{\lambda^+(\mathbf{k})}{\epsilon k}e^{-k(z-d/2)} + \frac{1}{2\epsilon k}\int_{d/2}^{\infty} dz'\,\rho(\mathbf{k},z')\left[e^{-k|z-z'|} + e^{-k(z+z'-d)}\right],\\ \Phi^-(\mathbf{k},z) &= \frac{\lambda^-(\mathbf{k})}{\epsilon k}e^{k(z+d/2)} + \frac{1}{2\epsilon k}\int_{-\infty}^{-d/2} dz'\,\rho(\mathbf{k},z')\left[e^{-k|z-z'|} + e^{k(z+z'+d)}\right].\end{aligned} \tag{S2.16}$$

Inside the membrane, the potential is

$$\Phi^m(\mathbf{k},z) = \frac{1}{2\epsilon_m k}\left[\frac{\cosh(kz)}{\sinh(kd/2)}(\lambda^+(\mathbf{k}) - \lambda^-(\mathbf{k})) + \frac{\sinh(kz)}{\cosh(kd/2)}(\lambda^+(\mathbf{k}) + \lambda^-(\mathbf{k}))\right]. \tag{S2.17}$$

The fields $\lambda^\pm$ are then determined by invoking continuity. For this, it is useful to define the quantities

$$\langle\rho^+(\mathbf{k})\rangle \coloneqq k\int_{d/2}^{\infty} dz'\,\rho(\mathbf{k},z')e^{-k(z-d/2)}, \quad \langle\rho^-(\mathbf{k})\rangle \coloneqq k\int_{-\infty}^{-d/2} dz'\,\rho(\mathbf{k},z')e^{k(z+d/2)}. \tag{S2.18}$$

These are exponentially-weighted averages of the charge density for each mode. Instead of characterizing the fields individually, it is more useful to work with the following quantities:

$$\begin{aligned}\lambda^+(\mathbf{k}) + \lambda^-(\mathbf{k}) &= \frac{1}{k}\left\{1 + \frac{\epsilon}{\epsilon_m}\tanh(kd/2)\right\}^{-1}(\langle\rho^+(\mathbf{k})\rangle - \langle\rho^-(\mathbf{k})\rangle),\\ \lambda^+(\mathbf{k}) - \lambda^-(\mathbf{k}) &= \frac{\epsilon_m}{\epsilon k}\frac{\tanh(kd/2)}{1 + \frac{\epsilon_m}{\epsilon}\tanh(kd/2)}(\langle\rho^+(\mathbf{k})\rangle + \langle\rho^-(\mathbf{k})\rangle).\end{aligned} \tag{S2.19}$$

This will allow us to obtain the appropriate boundary conditions for the charge dynamics.

S2.3 Charge profile boundary conditions

In the bulk the charge dynamics evolve according to Eq. (S2.9). We assume there is no free charge in the membrane and that ions cannot cross the boundary into the membrane. This enforces a no-flux condition on all ionic species:

$$j_i^\perp(\mathbf{x},z=0^+) = j_i^\perp(\mathbf{x},z=0^-) = 0, \tag{S2.20}$$

where $j_i^\perp$ is the component of the current perpendicular to the membrane. Here we have abused notation further by only considering z coordinates in the bulk, such that $z=0^+$ correspond to the top of the membrane and $z=0^-$ to the bottom. This simplifies the calculations, since the effects of the interior of the membrane only matter through the fields $\lambda^\pm$ in the rest of the analysis.

Using Eq. (S2.3) for the currents, the no-flux conditions translate to Neumann conditions for the ionic concentrations:

$$-D\partial_z n_i(\mathbf{x}, z=0^+) + \beta D n_i z_i e E^\perp(\mathbf{x}, z=0^+) = -D\partial_z n_i(\mathbf{x}, z=0^-) + \beta D n_i z_i e E^\perp(\mathbf{x}, z=0^-) = 0. \quad \text{(S2.21)}$$

Summing over species and transforming the $\mathbf{x}$ coordinates, we obtain analogous boundary conditions for the charge density:

$$\alpha E^\perp(\mathbf{k}, z=0^+) - D\partial_z \rho(\mathbf{k}, z=0^+) = \alpha E^\perp(\mathbf{k}, z=0^-) - D\partial_z \rho(\mathbf{k}, z=0^-) = 0. \quad \text{(S2.22)}$$

Using our solution for the potential we can get the electric fields at the membrane:

$$\begin{aligned} E^\perp(\mathbf{k}, z=0^+) &= -\partial_z \Phi^+(\mathbf{k}, z=0^+) = -\frac{\lambda^+(\mathbf{k})}{\epsilon}, \\ E^\perp(\mathbf{k}, z=0^-) &= -\partial_z \Phi^-(\mathbf{k}, z=0^-) = -\frac{\lambda^-(\mathbf{k})}{\epsilon}. \end{aligned} \quad \text{(S2.23)}$$

Now, the boundary conditions for the charge density can also be expressed using the following linear combinations:

$$\begin{aligned} \partial_z \rho(\mathbf{k}, z=0^+) + \partial_z \rho(\mathbf{k}, z=0^-) &= -\frac{\alpha}{\epsilon D}[\lambda^+(\mathbf{k}) + \lambda^-(\mathbf{k})], \\ \partial_z \rho(\mathbf{k}, z=0^+) - \partial_z \rho(\mathbf{k}, z=0^-) &= -\frac{\alpha}{\epsilon D}[\lambda^+(\mathbf{k}) - \lambda^-(\mathbf{k})]. \end{aligned} \quad \text{(S2.24)}$$

This is useful since an arbitrary profile $\rho(\mathbf{k}, z)$ can be decomposed into an odd and an even part, each of which satisfies one of these boundary conditions automatically. Explicitly:

$$\rho_{\text{odd}}(\mathbf{k}, z) = \frac{\rho(\mathbf{k}, z) - \rho(\mathbf{k}, -z)}{2}, \quad \rho_{\text{even}}(\mathbf{k}, z) = \frac{\rho(\mathbf{k}, z) + \rho(\mathbf{k}, -z)}{2}. \quad \text{(S2.25)}$$

In this decomposition, ρ_{odd} always satisfies the second boundary condition in Eq. (S2.24), while ρ_{even} always satisfies the first one. When characterizing measurements, we will focus on charge differences between the two sides of the membrane, which are only affected by ρ_{odd}. Therefore, from now on we only focus on odd profiles.

If our profile is odd, we can specify it for the half-interval $z > 0$ to obtain the full solution. Furthermore, we only need to have it satisfy the first boundary condition in Eq. (S2.24) to satisfy no-flux. Using Eq. (S2.19), our boundary condition can be written as (recall that $l_D^2 = D\epsilon/\alpha$)

$$\partial_z \rho(\mathbf{k}, z=0^+) = -\frac{1}{l_D^2}\left\{1 + \frac{\epsilon}{\epsilon_m}\tanh(kd/2)\right\}^{-1} \int_0^\infty dz'\, \rho(\mathbf{k}, z') e^{-kz'}. \quad \text{(S2.26)}$$

This boundary condition, along with the condition $\rho(\mathbf{k}, z \to \infty) \to 0$, will be sufficient to determine the structure of the dynamics of the charge profile.

S2.4 Mode structure of the charge dynamics

To simplify exposition, we will fix $\mathbf{k}$ throughout this subsection and omit the dependence of ρ on $\mathbf{k}$. Additionally, we define $\psi \coloneqq \left\{1 + \frac{\epsilon}{\epsilon_m}\tanh(kd/2)\right\}^{-1}$. Note that $\psi < 1$ and, for long wavelengths,

$$\psi \approx 1 - \frac{\epsilon}{2\epsilon_m} kd. \quad \text{(S2.27)}$$

We want to characterize the dynamics of the charge density profile, which follows the PDE

$$\tau_D \partial_t \rho = -(1 + k^2 l_D^2)\rho + l_D^2 \partial_z^2 \rho, \quad \text{(S2.28)}$$

subject to the boundary condition

$$\partial_z \rho(z=0^+) = -\frac{\psi}{l_D^2} \int_0^\infty dz'\, \rho(z') e^{-kz'}. \quad \text{(S2.29)}$$

To simplify the analysis, we can write the problem in terms of the following dimensionless variables:

$$s \coloneqq (1+k^2 l_D^2)\frac{t}{\tau_D}, \quad \zeta \coloneqq (1+k^2 l_D^2)^{1/2}\frac{z}{l_D}, \quad \kappa \coloneqq (1+k^2 l_D^2)^{-1/2} k l_D. \tag{S2.30}$$

Our PDE, then, becomes

$$\partial_s \rho = -\rho + \partial_\zeta^2 \rho, \tag{S2.31}$$

with corresponding boundary condition

$$\partial_\zeta \rho(\zeta = 0^+) = -\frac{\psi}{(1+kl_D^2)} \int_0^\infty d\zeta'\, \rho(\zeta') e^{-\kappa\zeta'}. \tag{S2.32}$$

We want to find the modes of the charge dynamics above, which are the eigenfunctions of the operator $-1+\partial_\zeta^2$ that satisfy the boundary condition. It is convenient to characterize these modes by their dimensionless relaxation time, which is the reciprocal of their corresponding eigenvalue. We denote with ϕ_{s_0} the mode with dimensionless relaxation time s_0. By definition, ϕ_{s_0} satisfies

$$-\frac{1}{s_0}\phi_{s_0}(\zeta) = -\phi_{s_0}(\zeta) + \phi''_{s_0}(\zeta). \tag{S2.33}$$

The general solution to this ODE is

$$\phi_{s_0}(\zeta) = A \exp\left[\left(\frac{s_0-1}{s_0}\right)^{1/2}\zeta\right] + B\exp\left[-\left(\frac{s_0-1}{s_0}\right)^{1/2}\zeta\right]. \tag{S2.34}$$

We can clearly see that there are two very different regions: when $s_0 > 1$ we have exponentially decaying modes (in space) and when $s_0 < 1$ we have oscillating modes. Let us go through these cases separately.

S2.4.1 Slow timescales ($s_0 > 1$)

Since we do not want exponentially increasing solutions, we must have $A = 0$, and can set $B = 1$. For this mode, the boundary condition translates to

$$\left(\frac{s_0-1}{s_0}\right)^{1/2} = \frac{\psi}{(1+kl_D^2)}\left\{\kappa + \left(\frac{s_0-1}{s_0}\right)^{1/2}\right\}^{-1}. \tag{S2.35}$$

This equation has a unique solution s_0^*. We can obtain the (dimensionful) decay timescale of this mode:

$$\tau_{\text{slow}} \coloneqq \frac{\tau_D}{(1+k^2 l_D^2)} s_0^*. \tag{S2.36}$$

For $k \ll l_D$, this decay time is

$$\tau_{\text{slow}} \approx \frac{2}{\alpha k}\left(\frac{1}{c_0} + \frac{2l_D}{\epsilon}\right)^{-1}, \tag{S2.37}$$

where $c_0 = \epsilon_m/d$ is the capacitance per unit area of the membrane. These is precisely the timescale of the coarse-grained 2D dynamics in [1], where the capacitance is replaced by an effective capacitance $\frac{1}{c_{\text{eff}}} = \frac{1}{c_0} + \frac{2l_D}{\epsilon}$.

The decay lengthscale for the slow mode is

$$\left(\frac{s_0^*}{s_0^*-1}\right)^{1/2}\frac{l_D}{(1_k^2 l_D^2)^{1/2}} \approx l_D. \tag{S2.38}$$

Since communication usually takes place at distances much larger than the Debye length, we see that the only long-lived mode that can carry information is this slow mode. All the other modes decay at a timescale smaller than τ_D in this regime.

S2.4.2 Fast timescales ($s_0 < 1$)

For this case, we can write our modes as

$$\phi_{s_0}(\zeta) = \sin\left[\left(\frac{1-s_0}{s_0}\right)^{1/2}\zeta\right] + Q(s_0)\cos\left[\left(\frac{1-s_0}{s_0}\right)^{1/2}\zeta\right]. \tag{S2.39}$$

The boundary condition takes the form

$$\left(\frac{1-s_0}{s_0}\right)^{1/2} = -\frac{\psi}{(1+k^2l_D^2)}\left[\frac{Q(s_0)\kappa + \left(\frac{1-s_0}{s_0}\right)^{1/2}}{\kappa^2 + \left(\frac{1-s_0}{s_0}\right)}\right]. \tag{S2.40}$$

This condition can always be satisfied by choosing

$$Q(s_0) = -\frac{1}{\kappa}\left(\frac{1-s_0}{s_0}\right)^{1/2}\left[\frac{\kappa^2 + \left(\frac{1-s_0}{s_0}\right)}{\psi/(1+k^2l_D^2)} + 1\right]. \tag{S2.41}$$

This means that there is a continuum of oscillatory "fast modes" which decay at a timescale shorter than the Debye time.

S3 Measurement fluctuations

S3.1 Coupling of the measurement to the free energy

We consider an ion channel that performs the following measurement of the charge in the bulk:

$$M(t) = \int d^2\mathbf{x}\, dz\, e^{-\|\mathbf{x}\|^2/2\sigma^2} e^{-|z|/h}\mathrm{sgn}(z)\rho(\mathbf{x}, z, t). \tag{S3.1}$$

Since this measurement is a functional of the charge density, we can consider a perturbation of the free energy that takes the same form as a coupling of the measurement to a fictitious field ζ:

$$\mathcal{F}[\mathbf{n}, \zeta] = \mathcal{F}_0[\mathbf{n}] - \zeta M. \tag{S3.2}$$

This coupling changes the dynamics of the model by altering the currents obtained from Eq. (S2.3). Let $\chi^{M\zeta}(\omega)$ denote the linear response kernel of the measurement to the coupling ζ, defined by

$$M(\omega) = \chi^{M\zeta}(\omega)\zeta(\omega). \tag{S3.3}$$

According to the fluctuation-dissipation theorem, the spectrum of equilibrium fluctuations of M satisfies

$$S^M(\omega) = -\frac{2k_BT}{\omega}\mathrm{Im}[\chi^{M\zeta}(\omega)]. \tag{S3.4}$$

Therefore, we can obtain the spectrum of the fluctuations by computing the response kernel $\chi^{M\zeta}(\omega)$. This is done in this section.

S3.2 Perturbed dynamics and boundary conditions

Let us define the kernel

$$\Gamma(\mathbf{r}) \coloneqq e^{-\|\mathbf{x}\|^2/2\sigma^2} e^{-|z|/h}\mathrm{sgn}(z). \tag{S3.5}$$

The ionic currents from the perturbed free energy can be written as

$$\mathbf{j}_i = -D\nabla n_i + \beta D n_i z_i e\mathbf{E} + \zeta\beta n_i z_i e\nabla\Gamma. \tag{S3.6}$$

Summing over species, we get the modified charge current

$$\mathbf{J} = \alpha\mathbf{E} - D\nabla\rho + \alpha\zeta\nabla\Gamma. \tag{S3.7}$$

Using conservation of chage, this leads to the modified dynamics

$$\tau_D\partial_t\rho = -\rho + l_D^2\nabla^2\rho + \epsilon\zeta\nabla^2\Gamma. \tag{S3.8}$$

The Laplacian of Γ is

$$\nabla^2\Gamma = \mathrm{sgn}(z)\left[\frac{1}{h^2} + \frac{\|\mathbf{x}\|^2}{\sigma^4} - \frac{2}{\sigma^2}\right] e^{-\|\mathbf{x}\|^2/2\sigma^2} e^{-|z|/h}. \tag{S3.9}$$

Therefore, taking the Fourier transform of the $\mathbf{x}$ coordinates, we obtain the following dynamics in Fourier space:

$$\tau_D\partial_t\rho = -(1 + k^2 l_D^2)\rho + l_D^2\partial_z^2\rho - 2\pi\epsilon\zeta\frac{\sigma^2}{h^2}(1 - k^2h^2)\mathrm{sgn}(z)e^{-k^2\sigma^2/2}e^{-|z|/h}. \tag{S3.10}$$

Going back to the modified currents, we see that the no-flux condition now reads

$$\partial_z\rho(\mathbf{x}, 0^\pm) = \frac{\alpha}{D}E^\perp(\mathbf{x}, 0^\pm) + \frac{\alpha\zeta}{D}\partial_z\Gamma(\mathbf{x}, 0^\pm). \tag{S3.11}$$

Evaluating the derivative of the kernel yields

$$\partial_z\Gamma(\mathbf{x}, 0^\pm) = -\frac{1}{h}e^{-\|\mathbf{x}\|^2/2\sigma^2}. \tag{S3.12}$$

This form of the perturbation only affects the odd part of the charge profile. Since M only depends on the odd part of the profile, we only need to find how this part responds to the coupling ζ. Therefore, we can focus on odd profiles that satisfy the following boundary condition:

$$\partial_z\rho(\mathbf{x}, 0^+) + \partial_z\rho(\mathbf{x}, 0^+) = \frac{\alpha}{D}(E^\perp(\mathbf{x}, 0^+) + E^\perp(\mathbf{x}, 0^-)) - 2\frac{\alpha\zeta}{Dh}e^{-\|\mathbf{x}\|^2/2\sigma^2}. \tag{S3.13}$$

Again using the solution for Poisson's equation, we can write the boundary condition for the solution on the half-plane $z > 0$ as

$$\partial_z\rho(\mathbf{k}, 0^+) = -\frac{\psi(\mathbf{k})}{l_D^2}\int_0^\infty dz\, e^{-kz}\rho(\mathbf{k}, z) - 2\pi\sigma^2\frac{\alpha\zeta}{Dh}e^{-k^2\sigma^2/2}. \tag{S3.14}$$

S3.3 The charge response kernel

To find the kernel $\chi^{M\zeta}(\omega)$, we need to first determine how $\rho(\mathbf{k}, \omega)$ responds to $\zeta(\omega)$. We begin by analyzing the bulk dynamics in frequency space. It is useful to define the following quantity:

$$q^2 \coloneqq 1 + k^2 l_D^2 + i\omega\tau_D. \tag{S3.15}$$

Then the dynamics can be written as

$$q^2\rho - l_D^2\partial_z^2\rho = -2\pi\epsilon\zeta\frac{\sigma^2}{h^2}(1 - k^2h^2)e^{-k^2\sigma^2/2}e^{-z/h}. \tag{S3.16}$$

We can to find the profile the solves this differential equation. Since we want solutions that do not grow exponentially as $z \to \infty$, the homogeneous solution is

$$\rho_h(z) = Ae^{-qz/l_D}. \tag{S3.17}$$

The particular solution is

$$\rho_p(z) = -2\pi\zeta\frac{\sigma^2\epsilon}{h^2}\frac{(1 - k^2h^2)}{(q^2 - l_D^2h^2)}e^{-k^2\sigma^2/2}e^{-z/h}. \tag{S3.18}$$

Our charge density is, then, $\rho = \rho_h + \rho_p$. To determine the coefficient A, we can use our boundary condition. It is useful to compute the following quantities:

$$\begin{aligned}\partial_z \rho(0^+) &= -\frac{Aq}{l_D} + 2\pi\zeta \frac{\sigma^2\epsilon}{h^3} \frac{(1-k^2h^2)}{(q^2 - l_D^2/h^2)} e^{-k^2\sigma^2/2}, \\ \int_0^\infty dz\, e^{-kz}\rho(z) &= \frac{A}{k+q/l_D} - 2\pi\zeta \frac{\sigma^2\epsilon}{h} \frac{(1-kh)}{(q^2 - l_D^2/h^2)} e^{-k^2\sigma^2/2}.\end{aligned} \tag{S3.19}$$

Plugging these into our boundary condition yields

$$\begin{aligned}&-\frac{Aq}{l_D} + 2\pi\zeta \frac{\sigma^2\epsilon}{h^3} \frac{(1-k^2h^2)}{(q^2 - l_D^2/h^2)} e^{-k^2\sigma^2/2} = \\ &-\frac{\psi}{l_D^2}\left[\frac{A}{k+q/l_D} - 2\pi\zeta \frac{\sigma^2\epsilon}{h} \frac{(1-kh)}{(q^2 - l_D^2/h^2)} e^{-k^2\sigma^2/2}\right] - 2\pi\zeta \frac{\sigma^2\epsilon}{hl_D^2} e^{-k^2\sigma^2/2}.\end{aligned} \tag{S3.20}$$

Let us define the following dimensionless quantities:

$$\kappa \coloneqq k l_D, \quad \eta \coloneqq \frac{h}{l_D}. \tag{S3.21}$$

Solving for A yields

$$A = -2\pi\zeta \frac{\sigma^2\epsilon}{l_D\eta} \underbrace{\left\{\frac{\psi}{\kappa+q} - q\right\}^{-1} \left[\frac{(1-\kappa^2\eta^2)}{(\eta^2q^2-1)} - \frac{\psi(1-\kappa\eta)}{(q^2-1/\eta^2)} + 1\right]}_{\coloneqq G_0}. \tag{S3.22}$$

Therefore, our solution to the charge density profile is

$$\rho(z,\omega) = -2\pi \frac{\sigma^2\epsilon}{l_D^2\eta^2}\left[\eta G_0 e^{-qz/l_D} + \frac{(1-\kappa^2\eta^2)}{(q^2-1/\eta^2)} e^{-z/h}\right] e^{-k^2\sigma^2/2}\zeta(\omega) \tag{S3.23}$$

This relationship allows to build the response function of the measurement.

S3.4 The response function

We can write the measurement in frequency space as

$$M(\omega) = \frac{\sigma^2}{2\pi}\int d^2\mathbf{k}\, dz\, e^{-k^2\sigma^2/2}\mathrm{sgn}(z) e^{-|z|/h}\rho(\mathbf{k}, z, \omega). \tag{S3.24}$$

We can use the response of ρ to ω to write this integral in terms of $\zeta(\omega)$:

$$M(\omega) = -2\frac{\sigma^4\epsilon}{l_D^2\eta^2}\left[\int d^2\mathbf{k}\, e^{-k^2\sigma^2}\int_0^\infty dz\, e^{-z/h}\left(\eta G_0 e^{-qz/l_D} + \frac{(1-\kappa^2\eta^2)}{(q^2-1/\eta^2)} e^{-z/h}\right)\right]\zeta(\omega) \tag{S3.25}$$

Evaluating the integral yields

$$M(\omega) = -2\frac{\sigma^4\epsilon}{l_D^2\eta^2}\left[\int d^2\mathbf{k}\, e^{-k^2\sigma^2}\left(\frac{\eta G_0}{q/l_D + 1/h} + \frac{l_D\eta}{2}\frac{(1-\kappa^2\eta^2)}{(q^2-1/\eta^2)}\right)\right]\zeta(\omega). \tag{S3.26}$$

Therefore, our response function is

$$\chi^{M\zeta}(\omega) = -\frac{\sigma^4\epsilon}{l_D}\int d^2\mathbf{k}\, e^{-k^2\sigma^2}\left(\frac{2G_0}{q\eta+1} + \eta\frac{(1-\kappa^2\eta^2)}{(q^2\eta^2-1)}\right). \tag{S3.27}$$

S3.5 The fluctuation spectrum

Using the fluctuation-dissipation theorem, we have that the spectrum of fluctuations of M is given by

$$S^M(\omega) = -\frac{2k_BT}{\omega}\mathrm{Im}[\chi^{M\zeta}(\omega)]. \tag{S3.28}$$

Now, we are working in a regime in which $\omega\tau_D$ is much smaller than all the other quantities in the problem. Let us define the following function:

$$G := \frac{2G_0}{q\eta+1} + \eta\frac{(1-\kappa^2\eta^2)}{(q^2\eta^2-1)} \tag{S3.29}$$

Note that ω only enters G through q. Therefore, to first order in ω, we can write

$$G(\omega) \approx G(\omega=0) + \left.\frac{\partial G}{\partial q}\right|_{\omega=0}\frac{i\omega\tau_D}{2q} \tag{S3.30}$$

where q is now evaluated at $\omega = 0$. This means that

$$\mathrm{Im}[G(\omega)] \approx \frac{\partial G}{\partial q}\frac{\omega\tau_D}{2q}. \tag{S3.31}$$

Using this in our expression for the fluctuations yields

$$S^M(\omega) \approx \frac{\sigma^4\epsilon\tau_D}{l_D}k_BT\int d^2\mathbf{k}\,\frac{e^{-k^2\sigma^2}}{q}\frac{\partial G}{\partial q} \tag{S3.32}$$

Since the integral is radially symmetric, we can write it as

$$S^M(\omega) \approx 2\pi\frac{\sigma^4\epsilon\tau_D}{l_D}k_BT\int_0^\infty dk\,k\frac{e^{-k^2\sigma^2}}{q}\frac{\partial G}{\partial q} \tag{S3.33}$$

In terms of the dimensionless variable κ, this is

$$S^M(\omega) \approx 2\pi\frac{\sigma^4\epsilon\tau_D}{l_D^3}k_BT\int_0^\infty d\kappa\,e^{-\kappa^2\sigma^2/l_D^2}\left[\frac{\kappa}{q}\frac{\partial G}{\partial q}\right] \tag{S3.34}$$

The integral in this expression is a complicated function of the sensor height h/l_D, sensor width σ/l_D and the other parameters of the problem. In the following section we study the problem of optimal design of sensors and characterize the scaling of noise for these.

S4 Optimal sensors and noise scaling

S4.1 Communication and sensing

Our analysis focuses on sensors in the context of a larger communication scheme, in which they are trying to sense signals from the environment. Often these signals originate from far away, and they need to be communicated using the electrical dynamics of the system. Following [1], we study a communication scheme in which a signal is generated at a distance $r \gg \sigma$ and is sensed by our sensor. We characterize the optimal sensing height for this problem.

Consider the communication scheme shown in Figure S1. A signal source located at a distance $r \gg \sigma$ from the sensor injects a time-dependent current $I(t)$. The output of the communication scheme is our measurement $M(t)$. As shown in [1], the information carried through this communication channel can be characterized via the following signal-to-noise ratio:

$$\mathrm{SNR}(\omega) = \frac{|\chi^{MI}(\omega)|^2 S^I(\omega)}{S^M(\omega)}, \tag{S4.1}$$

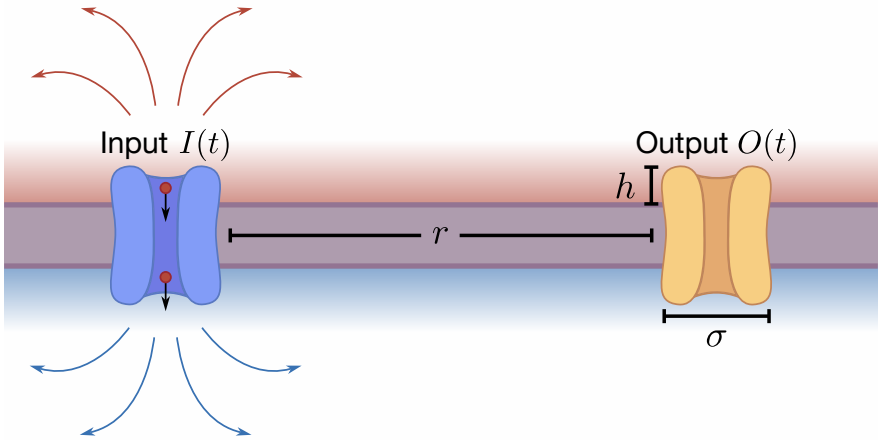


Figure S1: Communication scheme for two far away ion channels. The sender channel injects current at the membrane, while the output channel makes a local measurement of charge.

where $\chi^{MI}(\omega)$ is the linear response kernel of the measurement to the input and $S^I(\omega)$ is the power spectrum of the input.

The response kernel χ^{MI} depends on the dynamics of the charge density profile. For long-ranged communication, only the long-lived z mode in Section S2.4 can carry information. Furthermore, as shown in [1], the main contribution to the response kernel comes from $\mathbf{k}$ modes where kr is of order 1, which implies $kl_D \sim l_D/r \ll 1$. As we showed in Section S2.4, the long-lived z mode decays as $e^{-|z|/l_D}$ for $kl_D \ll 1$. This means the information that can be conveyed through electrical communication is confined to a layer of width l_D, which is called the Debye layer. Since the relevant information is contained within the Debye layer, we can isolate the h dependence of χ^{MI} as follows:

$$\chi^{MI}(\omega) \propto \int d^3\mathbf{r}\, \Gamma(\mathbf{r}) \mathrm{sgn}(z) e^{-|z|/l_D} \propto \left(\frac{h}{l_D + h}\right) = \left(\frac{\eta}{1+\eta}\right). \tag{S4.2}$$

Since $S^I(\omega)$ does not depend on height and we have the scaling of $\chi^{MI}(\omega)$ with h, we can now study how the signal-to-noise ratio behaves as a function of height, and how to use this to optimize our sensor.

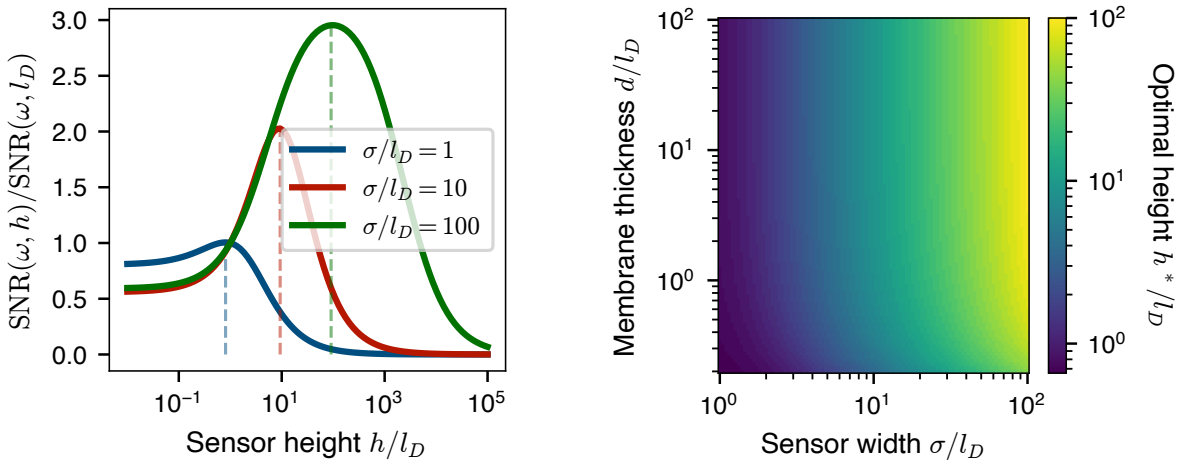
S4.2 Optimal sensor height

To obtain the optimal sensor height, we evaluate the signal-to-noise ratio in Eq. (S4.1) as a function of height for various values of sensor width σ and membrane height d. This is shown in Figure S2. We see that $\mathrm{SNR}(\omega, h)$ has a unique maximum for a wide range of parameters (Fig. S2.a), so there is a well-defined optimal sensor height. The optimal sensor height seems to mostly depend on σ rather than d (Fig. S2.b).

While it may seem that there is an advantage to having a very high sensor if σ is very large (~ 100 Debye lenghts), we see that this is because $\mathrm{SNR}(\omega, h)$ becomes very flat for high values of σ (Fig. S2.a). In fact, since noise increases with height and the signal χ^{MI} saturates at $h \sim l_D$, the sensor cannot benefit significantly from being more than a few Debye lengths high. Additionally, the fact that χ^{MI} saturates means that in order to have a flat signal-to-noise ratio, S^M must also be flat in this region. Therefore, even if the true optimal height can be large, a similar level of noise can be achieved with a sensor that is a few Debye lengths high.

S4.3 Scaling of fluctuations

We now have a numerical characterization of the optimal height of a sensor as a function of parameters. To gain further intuition, we would like to understand what drives the scaling of the noise. As we argued in



(a) Dependence of the signal-to-noise ratio on sensor height for the parameters in the main text.

(b) Optimal sensor height as a function of σ and d.

Figure S2: Optimization of the signal-to-noise ratio for $\epsilon/\epsilon_m = 20$.

S4.2, signals sent from far away are confined to the Debye layer, so l_D is the relevant length scale that should determine the scale of fluctuations. Therefore, we will analytically study the behavior of fluctuations for a sensor of height $h = l_D$.

Since we are interested in studying sensors that operate at length scales larger than l_D, we analyze what happens when we expand the fluctuations (Eq. (S3.34)) in l_D/σ. We can do this by expanding the following integrand in κ:

$$\frac{\kappa}{q}\frac{\partial G}{\partial q} = a_0 + a_1\kappa + \dots \tag{S4.3}$$

This gives an expansion of the following integral:

$$\int_0^\infty d\kappa\, e^{-\kappa^2\tilde{\sigma}^2}\left[\frac{\kappa}{q}\frac{\partial G}{\partial q}\right] = \tilde{a}_0\left(\frac{l_D}{\sigma}\right) + \tilde{a}_1\left(\frac{l_D}{\sigma}\right)^2 + \dots, \tag{S4.4}$$

where $\tilde{a}_n = a_n\int_0^\infty dx\, x^n e^{-x^2}$. Finally, we can use this to obtain an expansion of the fluctuations in l_D/σ:

$$S^M(\omega) = S_0^M(\omega) + S_1^M(\omega) + \dots \tag{S4.5}$$

By evaluating the terms in this expansion, we can gain an intuition for the main sources of noise in our system.

The first two coefficients in our expansion of the integrand are

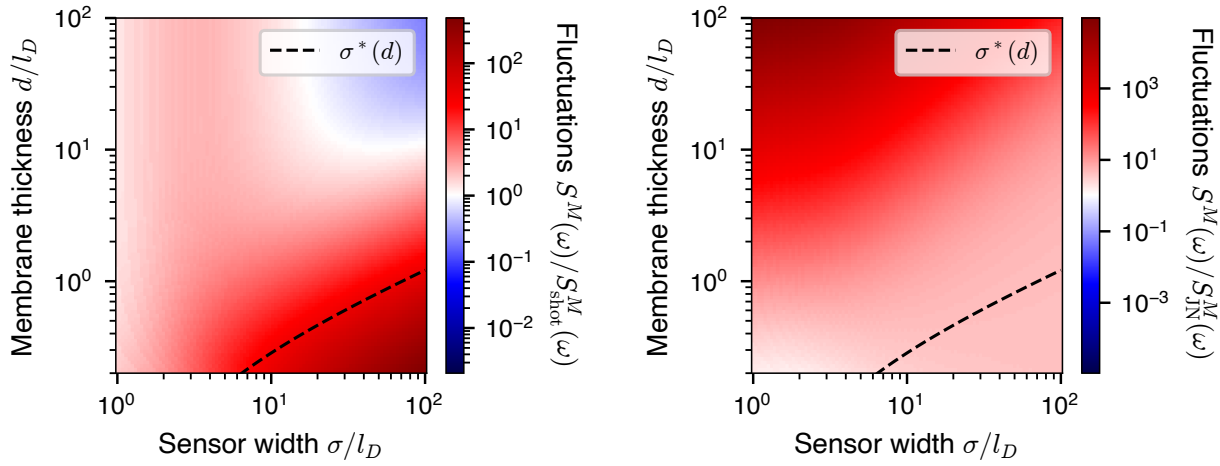
$$a_0 = \left(1 + \frac{1}{2}\frac{\epsilon}{\epsilon_m}\frac{d}{l_D}\right)^{-2} = \frac{4l_D^2 c_{\text{eff}}^2}{\epsilon^2}, \quad a_1 = \frac{1}{4}\left[1 + \frac{3\frac{\epsilon d}{\epsilon_m l_D}}{\left(1+\frac{\epsilon d}{2\epsilon_m l_D}\right)^2}\right], \tag{S4.6}$$

Plugging this into the expression for the fluctuation spectrum, we get an expansion of the form

$$S^M(\omega) = S_0^M(\omega) + S_1^M(\omega) + \dots \tag{S4.7}$$

With out expansion above, the leading terms in the fluctuation expansion are

$$S_0^M(\omega) = 4\pi^{3/2}\frac{\sigma^3\tau_D c_{\text{eff}}^2}{\epsilon}k_BT, \quad S_1^M(\omega) = \left[1 + \frac{3\frac{\epsilon d}{\epsilon_m l_D}}{\left(1+\frac{\epsilon d}{2\epsilon_m l_D}\right)^2}\right]\frac{\pi}{4}\frac{\sigma^2\epsilon\tau_D}{l_D}k_BT. \tag{S4.8}$$



(a) Comparison of fluctuations to shot noise.

(b) Comparison of fluctuations to Johnson-Nyquist noise.

Figure S3: Scaling of fluctuations for $\epsilon/\epsilon_m = 20$. The dashed line represents the values of (σ, d) at which shot noise and Johnson-Nyquist noise coincide.

From this expansion we can define two natural scales of noise:

$$S^M_{\rm JN}(\omega) := 4\pi^{3/2}\frac{\sigma^3 c_{\rm eff}^2}{\alpha}k_BT, \quad S^M_{\rm shot}(\omega) := \frac{\pi}{4}\frac{\sigma^2\epsilon\tau_D}{l_D}k_BT, \tag{S4.9}$$

where the effective capacitance is $\frac{1}{c_{\rm eff}} = \frac{1}{c_0} + \frac{2l_D}{\epsilon}$. The term $S^M_{\rm JN}(\omega)$ corresponds to Johnson-Nyquist noise, which is the thermal noise obtained in a continuum model of charge dynamics [1]. We can identify the second term with shot noise, which depends only on the local fluctuations of the charge density, and is independent of the capacitance of the system. We use this expression to analyze $S^M_1(\omega)$ since the presence of the membrane does not significantly change the magnitude of this contribution:

$$S^M_{\rm shot}(\omega) \le S^M_1(\omega) \le \frac{5}{2}S^M_{\rm shot}(\omega). \tag{S4.10}$$

In contrast, the Johnson-Nyquist term depends strongly on the capacitance of the membrane. For the parameters in the main text we have $c_0 \approx 10^{-2}\epsilon/l_D$, meaning that the effective capacitance is dominated by the membrane capacitance, so the Johnson-Nyquist fluctuations are heavily suppressed. We can evaluate when we would expect the shot noise contribution to dominate by evaluating at what value of σ we have $S^M_{\rm JN}(\omega) = S^M_{\rm shot}(\omega)$. This value is given by

$$\frac{\sigma^*(d)}{l_D} = \frac{2}{\pi^{1/2}}\left(1 + \frac{\epsilon d}{2\epsilon_m l_D}\right)^2 \tag{S4.11}$$

Therefore, for $\sigma > \sigma^*(d)$, the system is the Johnson-Nyquist regime, and a continuum description of the charge dynamics becomes appropriate.

S5 Sensing with a charged plate

S5.1 Setup

We now consider an alternate sensing mechanism in which the electric field, rather than the charge, is sensed. We consider a sensor composed of two plates, one with charge Q and another of charge $-Q$. They both have a width σ and are embedded inside the membrane. The displacement of the positive plate in the z direction is ξ, and we impose that the negative plate has a displacement $-\xi$. We assume that there plate is confined

mechanically by a quadratic potential, in addition to its interaction with the ionic charges. Therefore, the free energy of the system in the presence of the plate is

$$\mathcal{F} = \mathcal{F}_0 + \frac{K}{2}\xi^2 + \frac{Q}{2\pi\sigma^2}\int d^2\mathbf{x}\, e^{-\|\mathbf{x}\|^2/2\sigma^2}[\Phi_m^{\text{ions}}(\mathbf{x}, z=\xi) - \Phi_m^{\text{ions}}(\mathbf{x}, z=-\xi)], \tag{S5.1}$$

where $\mathcal{F}_0$ is the free energy in Eq. (S2.1) and Φ^{ions} is the potential generated by the ions.

We assume that the plate is overdamped and has mobility μ, such that its dynamics are dictated by

$$\partial_t \xi = -\mu\left(\frac{\partial \mathcal{F}}{\partial \xi}\right). \tag{S5.2}$$

With this specification, we can study the equilibrium fluctuations in ξ, which we take to be the sensor's measurement. These fluctuations determine the ability of the sensor to discriminate changes in voltage.

S5.2 Electric potential

The movement of the plate causes movement in the ionic charge near the membrane, which then feeds back into its dynamics. Therefore, to begin our study of the dynamics of the plate, we need to specify how the presence of the plate affects the distribution of ions.

Let us consider a setup with a fixed ionic charge distribution ρ and plate position ξ. The electric potential can be obtained as the sum of the potentials generated by the ionic charges and the plates, each satisfying the boundary conditions in Eqs. (S2.13) and (S2.14). Thus, we can use our solution in Eqs. (S2.16) and (S2.17) for Φ^{ions}.

We can first solve for the potential generated by the positive plate. For this, we need to solve Poisson's equation:

$$(\partial_z^2 - k^2)\Phi^{\text{plate},+}(\mathbf{k}, z) = -\frac{Q}{\epsilon_m}e^{-k^2\sigma^2/2}\delta(z-\xi). \tag{S5.3}$$

In the bulk, we have exponentially-decaying solutions:

$$\Phi_{\pm}^{\text{plate}}(\mathbf{k}, z) = A_{\pm}e^{-(|z|-d/2)}. \tag{S5.4}$$

Inside the plate, we can use the following bases, which satisfy the boundary conditions on the top and bottom interface, respectively,

$$\begin{aligned} L_+(z) &= \cosh(k(z-d/2)) - \frac{\epsilon}{\epsilon_m}\sinh(k(z-d/2)), \\ L_-(z) &= \cosh(k(z+d/2)) + \frac{\epsilon}{\epsilon_m}\sinh(k(z+d/2)). \end{aligned} \tag{S5.5}$$

Our solution inside the membrane, then, can be written as

$$\Phi_m^{\text{plate}}(\mathbf{k}, z) = \begin{cases} A_+L_+(z) & \xi < z < d/2, \\ A_-L_-(z) & -d/2 < z < \xi. \end{cases} \tag{S5.6}$$

At the location of the plate ($z = \xi$), we have

$$\partial_z\Phi_m^{\text{plate},+}(\mathbf{k}, \xi^+) - \partial_z\Phi_m^{\text{plate},+}(\mathbf{k}, \xi^-) = -\frac{Q}{\epsilon_m}e^{-k^2\sigma^2/2}. \tag{S5.7}$$

Using this condition and continuity, we obtain the solution for the potential generated by the plate:

$$\Phi_m^{\text{plate},+}(\mathbf{k}, z) = \frac{Qe^{-k^2\sigma^2/2}}{\epsilon_m k\Delta}\begin{cases} L_-(\xi)L_+(z) & \xi < z < d/2, \\ L_+(\xi)L_-(z) & -d/2 < z < \xi, \end{cases} \tag{S5.8}$$

where

$$\Delta \coloneqq \left(1+\frac{\epsilon^2}{\epsilon_m^2}\right)\sinh(kd)+\frac{2\epsilon}{\epsilon_m}\cosh(kd). \tag{S5.9}$$

To obtain the total potential generated by the sensor, we can use the symmetry of the problem and add both potentials:

$$\Phi^{\text{plate}}(\mathbf{k},z)=\Phi^{\text{plate},+}(\mathbf{k},z)+\Phi^{\text{plate},-}(\mathbf{k},z)=\Phi^{\text{plate},+}(\mathbf{k},z)-\Phi^{\text{plate},+}(\mathbf{k},-z). \tag{S5.10}$$

S5.3 Charge dynamics

In this section, we take z to specify coordinates in the bulk for the sake of simplicity. Thus, $z=0^{\pm}$ now denotes the bulk right above and below the membrane.

Since no charges are being introduced into the bulk, the charge dynamics we derived for our PNP analysis still hold, meaning that the evolution of charge is dictated by

$$\tau_D\partial_t\rho=-(1+k^2l_D^2)\rho+l_D^2\partial_z^2\rho. \tag{S5.11}$$

Our no-flux condition still reads

$$\partial_z\rho(\mathbf{k},z=0^{\pm})=\frac{\alpha}{D}E_z(\mathbf{k},z=0^{\pm}). \tag{S5.12}$$

What changes in the presence of the plate is that the electric field now has a contribution from the plates. We will eventually be interested in the linear response regime for ξ, so here we characterize how the charges respond to a small perturbation in ξ. The field generated by the plates on either side of the membrane is

$$E_z^{\text{plate}}(\mathbf{k},0^{\pm})=-\partial_z\Phi_z^{\text{plate}}(\mathbf{k},0^{\pm})=\frac{Qe^{-k^2\sigma^2/2}}{\epsilon_m\Delta}[L_-(\xi)-L_+(\xi)]. \tag{S5.13}$$

For small ξ, this is

$$E_z^{\text{plate}}(\mathbf{k},0^{\pm})=\frac{Qe^{-k^2\sigma^2/2}}{\epsilon_m}\left(\frac{k\xi}{C+(\epsilon/\epsilon_m)S}\right)+\mathcal{O}(\xi^2), \tag{S5.14}$$

where $C\coloneqq\cosh(kd/2)$ and $S\coloneqq\sinh(kd/2)$.

Given the form of the electric field and our charge boundary conditions, we see that changing ξ does not cause a change in the even component of the field, only on the odd one. Thus, we can focus on ρ_{odd} for our response analysis. Using our previous characterization of the electric field generated by the ionic charges, we can write our odd boundary condition as

$$\partial_z\rho_{\text{odd}}(\mathbf{k},0^+)=-\frac{\psi}{l_D^2}\int_0^\infty dz\,e^{-kz}\rho_{\text{odd}}(\mathbf{k},z)+\frac{\alpha}{D}\frac{Qe^{-k^2\sigma^2/2}}{\epsilon_m[C+(\epsilon/\epsilon_m)S]}k\xi. \tag{S5.15}$$

Now, in the frequency domain, our dynamics take the form

$$l_D^2\partial_z^2\rho=(1+k^2l_D^2+i\omega\tau_D)\rho=:q^2\rho. \tag{S5.16}$$

Thus, our charge distribution is given by

$$\rho(\mathbf{k},z)=Be^{-qz/l_D}. \tag{S5.17}$$

Using this ansatz in our boundary condition and solving for B, we obtain

$$B=\frac{\epsilon}{l_D}\left[\frac{\psi}{q+kl_D}-q\right]^{-1}\frac{Qe^{-k^2\sigma^2/2}}{\epsilon_m[C+(\epsilon/\epsilon_m)S]}k\xi. \tag{S5.18}$$

This allows us to characterize the response of the charge distribution to changes in the plate position. We can now use this to study the full response of the system to perturbations.

S5.4 Response to perturbations

To characterize the equilibrium fluctuations of ξ, we study the response of the system to a perturbation of the form

$$\mathcal{F} \to \mathcal{F} - r\xi. \tag{S5.19}$$

The dynamics of the sensor under this perturbation are

$$\mu^{-1}\partial_t \xi = -K\xi - \frac{Q}{2\pi\sigma^2}\int d^2\mathbf{x}\, e^{-\|\mathbf{x}\|^2/2\sigma^2}[\partial_z \Phi_m^{\text{ions}}(\mathbf{x},\xi) + \partial_z \Phi_m^{\text{ions}}(\mathbf{x},-\xi)] + r. \tag{S5.20}$$

Note that we can close these dynamics using our characterization of the electric field. For small perturbations ξ, we have

$$E_z^{\text{ions}}(\mathbf{k},\xi) = -\frac{(\lambda^+(\mathbf{k}) + \lambda^-(\mathbf{k}))}{2\epsilon_m C} - \frac{(\lambda^+(\mathbf{k}) - \lambda^-(\mathbf{k}))}{2\epsilon_m S} k\xi + \mathcal{O}(\xi^2), \tag{S5.21}$$

where $\lambda^\pm$ are defined in Eq. (S2.19). Thus, writing our dynamics in terms of a $\mathbf{k}$ integral, we obtain

$$\mu^{-1}\partial_t \xi = -K\xi - \frac{Q}{4\pi^2\epsilon_m}\int d^2\mathbf{k}\, e^{-k^2\sigma^2/2}\frac{(\lambda^+(\mathbf{k}) + \lambda^-(\mathbf{k}))}{C} + r. \tag{S5.22}$$

The ionic interaction integral can be written in terms of the odd charge distribution:

$$\lambda^+(\mathbf{k}) + \lambda^-(\mathbf{k}) = 2\psi\int_0^\infty dz\, \rho_{\text{odd}}(\mathbf{k},z)e^{-kz}. \tag{S5.23}$$

Using our characterization in the frequency domain, this is

$$\lambda^+(\mathbf{k}) + \lambda^-(\mathbf{k}) = \frac{2\psi B}{k + q/l_D} = 2\frac{\epsilon}{\epsilon_m}Q\left(\frac{\psi}{\psi - q(q+kl_D)}\right)\frac{e^{-k^2\sigma^2/2}k\xi(\omega)}{C + (\epsilon/\epsilon_m)S}. \tag{S5.24}$$

Define the effective electric spring constant as

$$K_{\text{el}} \coloneqq \frac{Q^2}{\pi\epsilon_m}\frac{\epsilon}{\epsilon_m}\int_0^\infty dk\, e^{-k^2\sigma^2}\frac{k^2}{C(C + (\epsilon/\epsilon_m)S)}\left(\frac{\psi}{\psi - q(q+kl_D)}\right). \tag{S5.25}$$

With this, the dynamics of the sensor in frequency space are given by

$$i\omega\mu^{-1}\xi(\omega) = -(K + K_{\text{el}})\xi(\omega) + r(\omega). \tag{S5.26}$$

The response function to perturbations, then, is

$$\chi^{\xi r}(\omega) = \frac{\mu}{\mu(K + K_{\text{el}}) + i\omega}. \tag{S5.27}$$

S5.5 The fluctuation spectrum

Using the fluctuation-dissipation theorem, we have that the spectrum of ξ fluctuations is

$$S^\xi(\omega) = -\frac{2k_BT}{\omega}\text{Im}[\chi^{\xi r}(\omega)]. \tag{S5.28}$$

If $\omega\mu K, \omega\tau_D \ll 1$, then we can approximate the spectrum to leading order in ω;

$$S^\xi(\omega) \approx \frac{2k_BT}{\mu(K + K_{\text{el}})^2} + \frac{2k_BT}{(K + K_{\text{el}})^2}\left.\frac{\partial K_{\text{el}}}{\partial(i\omega)}\right|_{\omega=0}. \tag{S5.29}$$

To compute this last derivative, it is useful to write $K_{\rm el}$ in terms of a dimensionless integral:

$$\begin{aligned} K_{\rm el} &= \frac{Q^2}{\pi\epsilon_m l_D^3}\frac{\epsilon}{\epsilon_m}\int_0^\infty d\kappa\, e^{-\kappa^2(\sigma^2/l_D^2)}\frac{\kappa^2}{C(C+(\epsilon/\epsilon_m)S)}\left(\frac{\psi}{\psi - q(q+\kappa)}\right) \\ &=: \frac{Q^2}{\pi\epsilon_m l_D^3}\frac{\epsilon}{\epsilon_m}\int_0^\infty d\kappa\, e^{-\kappa^2(\sigma^2/l_D^2)}\mathcal{I}(\kappa,q). \end{aligned} \tag{S5.30}$$

Since ω only enters $K_{\rm el}$ through q, we can write our derivative as

$$\left.\frac{\partial K_{\rm el}}{\partial(i\omega)}\right|_{\omega=0} = \frac{\tau_D}{2}\frac{Q^2}{\pi\epsilon_m l_D^3}\frac{\epsilon}{\epsilon_m}\int_0^\infty d\kappa\, e^{-\kappa^2(\sigma^2/l_D^2)}\frac{1}{q}\left.\frac{\partial\mathcal{I}}{\partial q}\right|_{q=(1+\kappa^2)^{1/2}}. \tag{S5.31}$$

To make progress in analyzing our fluctuation spectrum, we can analyze the large σ/l_D limit, as before. To do this, we expand our integrand around $\kappa = 0$. The first two leading terms are

$$\left.\frac{1}{q}\frac{\partial\mathcal{I}}{\partial q}\right|_{q=(1+\kappa^2)^{1/2}} = \frac{8}{\left(\frac{\epsilon}{\epsilon_m}\frac{d}{l_D}+2\right)^2}\left[1-\frac{3}{2}\kappa\right] + \mathcal{O}(\kappa^2). \tag{S5.32}$$

This give the following expansion of our fluctuation spectrum:

$$S^\xi(\omega) \approx \frac{2k_BT}{\mu(K+K_{\rm el})^2} + \frac{k_BT}{(K+K_{\rm el})^2}\frac{4\pi^{-1/2}Q^2\tau_D}{l_D^3\left(\frac{\epsilon}{\epsilon_m}\frac{d}{l_D}+2\right)^2}\frac{\epsilon}{\epsilon_m^2}\frac{l_D}{\sigma}\left[1-\frac{3}{2\pi^{1/2}}\frac{l_D}{\sigma}\right] + \mathcal{O}\left(\left(\frac{\sigma}{l_D}\right)^{-3}\right). \tag{S5.33}$$

We can write this expression in terms of the effective capacitance of the membrane:

$$S^\xi(\omega) \approx \frac{2k_BT}{\mu(K+K_{\rm el})^2} + 4\pi^{-1/2}\frac{k_BT}{(K+K_{\rm el})^2}\frac{Q^2\tau_D c_{\rm eff}^2}{\sigma\epsilon_m^2\epsilon}\left[1-\frac{3}{2\pi^{1/2}}\frac{l_D}{\sigma}\right] + \mathcal{O}\left(\left(\frac{\sigma}{l_D}\right)^{-3}\right). \tag{S5.34}$$

We are interested in the fluctuations in a measurement of the voltage across the membrane. Assuming the mechanical spring dominates the effective ionic potential for low frequencies ($K \gg K_{\rm el}(\omega\to 0)$), we can do a simple force balance to relate the measured potential to the spring displacement:

$$\frac{QV}{d} \approx K\xi. \tag{S5.35}$$

The fluctuations in the measurement of V, then, satisfy

$$S^V_{\rm plate}(\omega) \approx \frac{K^2d^2}{Q^2}S^\xi(\omega). \tag{S5.36}$$

Using our result for the fluctuations in ξ and using $K \gg K_{\rm el}$, we obtain

$$S^V_{\rm plate}(\omega) \approx \frac{2d^2k_BT}{\mu Q^2} + \frac{4\pi^{-1/2}}{\alpha\sigma}\frac{c_{\rm eff}^2}{c_0^2}k_BT\left[1-\frac{3}{2\pi^{1/2}}\frac{l_D}{\sigma}\right]. \tag{S5.37}$$

In the regime we are working in, we have $c_{\rm eff}\approx c_0$, so our fluctuations are

$$S^V_{\rm plate}(\omega) \approx \frac{2d^2k_BT}{\mu Q^2} + \frac{4\pi^{-1/2}}{\alpha\sigma}k_BT\left[1-\frac{3}{2\pi^{1/2}}\frac{l_D}{\sigma}\right] =: S^V_{\rm plate,0}(\omega) + S^V_{\rm plate,1}(\omega). \tag{S5.38}$$

S5.6 Comparison to charge measurement

In our bulk charge measurement calculation, the spectrum of fluctuations could be approximated by a Johnson-Nyquist noise and a shot noise contribution:

$$S^M_{\rm bulk}(\omega) \approx S^M_{\rm JN}(\omega) + S^M_{\rm shot}(\omega). \tag{S5.39}$$

We can translate these fluctuations to fluctuations in a measurement of the potential by looking at the estimate $V \approx M/2\pi\sigma^2 c_{\text{eff}}$. Thus, fluctuations in voltage measurements inferred from the bulk have the following structure:

$$S^V_{\text{bulk}}(\omega) = \frac{\pi^{-1/2}}{\sigma\alpha} k_B T + \frac{1}{16\pi} \frac{\epsilon \tau_D}{\sigma^2 l_D c_{\text{eff}}^2} k_B T. \tag{S5.40}$$

Note that the Johnson-Nyquist contribution has the same scaling as the second term of S^V_{plate}. Intuitively, this is because Johnson-Nyquist noise comes from the stable modes used for communication. In contrast, the other sources of noise are qualitatively different in both schemes, arising from sensor implementation details. To compare these, it is useful to write the shot noise contribution in terms of the ion mobility $\mu_{\text{ions}} = \beta D$. With this, the ratio between the fluctuation spectra is

$$\frac{S^V_{\text{shot}}(\omega)}{S^V_{\text{plate},0}(\omega)} = \frac{1}{32\pi} \frac{\epsilon^2}{\epsilon_m^2} \frac{Q^2}{\sigma^2 l_D \sum_i n_i^0 e^2 z_i^2} \frac{\mu}{\mu_{\text{ions}}}. \tag{S5.41}$$

The relation between these fluctuations can be made more interpretable by defining the average square charge in the bulk sensor:

$$Q^2_{\text{bulk}} \coloneqq \sigma^2 l_D \sum_i n_i^0 e^2 z_i^2. \tag{S5.42}$$

Thus, we can write the ratio as

$$\frac{S^V_{\text{shot}}(\omega)}{S^V_{\text{plate},0}(\omega)} = \frac{1}{32\pi} \frac{\epsilon^2}{\epsilon_m^2} \frac{Q^2}{Q^2_{\text{bulk}}} \frac{\mu}{\mu_{\text{ions}}}. \tag{S5.43}$$

This characterization makes it clear that the first source of noise in the plate's measurements is of the same nature as our bulk shot noise. The bulk permittivity is around an order of magnitude higher than the membrane permittivity. Ion channels would need to implement an electric field sensor using charged residues, meaning that Q is of the order of a few elementary charges and that the mobility μ cannot be much higher than the mobility of individual ions. This implies that this source of noise is a manifestation of shot noise in the plate's measurement arising from the interaction of its discrete components and the environment.

S6 Collective sensing

S6.1 Input signal and measurements

We now study a continuum model of electrical communication on a spherical neuron. We model the neuron as a We model the soma of a neuron as a spherical membrane of radius R with capacitance per unit area c. As before, it is embedded in a bulk medium with conductivity α. The charge density on the membrane is $\lambda(\mathbf{x})$, where $\mathbf{x}$ now represent coordinates on the surface of the sphere, such that $\|\mathbf{x}\| = R$.

To model communication, we consider a "sender" located at the north pole of the sphere that injects a time-dependent electrical current $I(t)$ spread over an area of linear size σ_I. To capture this, we model the input current density $J(\mathbf{x}, t)$ using a von Mises-Fisher distribution:

$$J(\mathbf{x}, t) = I(t) \frac{e^{(\mathbf{x} \cdot R\hat{\mathbf{z}})/\sigma_I^2}}{4\pi\sigma_I^2 \sinh(R^2/\sigma_I^2)}. \tag{S6.1}$$

The total current over the sphere is

$$\int ds(\mathbf{x})\, J(\mathbf{x}, t) = I(t), \tag{S6.2}$$

where $ds(\mathbf{x}) \coloneqq R^2 \sin(\theta)\, d\theta\, d\phi$ is the area element. In the $\sigma_I/R \to 0$ limit, where the system approximates an infinite plane, we recover a Gaussian spread of the current:

$$J(\mathbf{x}, t) \approx I(t) \frac{e^{(\mathbf{x} \cdot R\hat{\mathbf{z}} - R^2)/\sigma_I^2}}{2\pi\sigma_I^2} \approx I(t) \frac{e^{-\|x_\perp\|^2/2\sigma_I^2}}{2\pi\sigma_I^2}, \tag{S6.3}$$

where $\mathbf{x}_\perp$ is the component of $\mathbf{x}$ perpendicular to $\hat{\mathbf{z}}$.

Motivated by the von Mises-Fisher distribution, we define the input and measurement Gaussian distance factors between two vectors $\mathbf{x}$ and $\mathbf{y}$ as

$$\gamma^I(\mathbf{x},\mathbf{y}) \coloneqq e^{(\mathbf{x}\cdot\mathbf{y}-R^2)/\sigma_I^2}, \quad \gamma^M(\mathbf{x},\mathbf{y}) \coloneqq e^{(\mathbf{x}\cdot\mathbf{y}-R^2)/\sigma^2}. \tag{S6.4}$$

The input current can be written in terms of this distance factor as

$$J(\mathbf{x},t) \approx I(t)\frac{\gamma^I(\mathbf{x},R\hat{\mathbf{z}})}{2\pi\sigma_I^2}. \tag{S6.5}$$

We consider two communication schemes. In the first, which we call the "perfect instrument", the charge over the entire cell is measured, so the output is $M_{\text{cell}}(t) = \int ds(\mathbf{x})\,\lambda(\mathbf{x},t)$. The output of our second communication scheme is a vector of measurements $M_1(t),\ldots,M_N(t)$ made by N ion channels at locations $\mathbf{y}_1,\ldots,\mathbf{y}_N$. The measurement of ion channel j measures the charge in an area of linear size σ around its position $\mathbf{y}_j$. In terms of our spherical distance, this is

$$M_j(t) = \int ds(\mathbf{x})\,\gamma^M(\mathbf{x},\mathbf{y}_j)\lambda(\mathbf{x}). \tag{S6.6}$$

Throughout the rest of the analysis, we assume that $\sigma,\sigma_I \ll R$.

S6.2 Continuum charge dynamics

We now derive the dynamics in our continuum model of charge dynamics. We begin by analyzing the dynamics in the absence of noise.

The potential difference between the inside and outside of the membrane is

$$\Phi_+(\mathbf{x},t) - \Phi_-(\mathbf{x},t) = \frac{\lambda(\mathbf{x},t)}{c}, \tag{S6.7}$$

where Φ_+ and Φ_- represent the potential in the bulk outside and inside the membrane, respectively. The harmonic decomposition of Φ_+ and Φ_- is

$$\Phi_\pm(r,\theta,\phi) = \sum_{\ell,m}\Phi^\pm_{\ell m}(r)Y^m_\ell(\theta,\phi). \tag{S6.8}$$

In the bulk, the potential must satisfy Laplace's equation ($\nabla^2\Phi = 0$). In spherical coordinates this is

$$\nabla^2\Phi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\Phi}{\partial r}\right) + \frac{1}{r^2\sin(\theta)}\frac{\partial}{\partial\theta}\left(\sin(\theta)\frac{\partial\Phi}{\partial\theta}\right) + \frac{1}{r^2\sin^2(\theta)}\frac{\partial^2\Phi}{\partial\phi^2} = 0. \tag{S6.9}$$

This equation has solutions for the harmonic components of the form

$$\Phi^\pm_{\ell m}(r) = A^\pm_{\ell m}r^\ell + B^\pm_{\ell m}r^{-(\ell+1)}. \tag{S6.10}$$

Our first set of boundary conditions are that Φ must vanish as $r\to\infty$ and it must remain finite as $r\to 0$. This implies that $A^+_{\ell m} = B^-_{\ell m} = 0$. To characterize the rest of the solution, we study how the bulk currents affect the charge density dynamics.

Bulk currents are given by Ohm's law: $\mathbf{J}_{\text{bulk}}(\mathbf{x},t) = -\alpha\nabla\Phi$. At the membrane surface, charge accumulates due to bulk currents and drains from leak currents:

$$\partial_t\lambda(\mathbf{x},t) = \alpha\left.\frac{\partial\Phi_+(r,\mathbf{x},t)}{\partial r}\right|_{r=R} + J_{\text{leak}}(\mathbf{x},t) = \alpha\left.\frac{\partial\Phi_-(r,\mathbf{x},t)}{\partial r}\right|_{r=R} + J_{\text{leak}}(\mathbf{x},t), \tag{S6.11}$$

where the last equality is obtained by imposing that the charge inside the membrane has the same magnitude and opposite sign as the charge outside. Transforming this equation, we obtain an additional restriction on the coefficients:

$$\left.\frac{\partial\Phi^+_{\ell m}(r)}{\partial r}\right|_{r=R} = \left.\frac{\partial\Phi^-_{\ell m}(r)}{\partial r}\right|_{r=R}. \tag{S6.12}$$

With this, the solution is fully specified. Using $J_{\text{leak}}(\mathbf{x},t) = -\lambda(\mathbf{x},t)/\tau_{\text{leak}}$, the dynamics of the charge density field are

$$\partial_t \lambda_\ell^m(t) = -\frac{1}{\tau_\ell}\lambda_\ell^m(t), \tag{S6.13}$$

where the RC time of the ℓ-th harmonic satisfies

$$\tau_\ell^{-1} = \tau_{\text{leak}}^{-1} + \frac{\alpha}{Rc}\frac{\ell(\ell+1)}{2\ell+1}. \tag{S6.14}$$

For the parameters on the main text, we have $\alpha\tau_{\text{leak}}/Rc \approx 10^4$. Therefore, we have $\tau_\ell = \tau_{\text{leak}}$ for $\ell = 0$ and $\tau_\ell \approx \frac{Rc}{\alpha}\frac{2\ell+1}{\ell(\ell+1)}$ for all $\ell > 0$. As we show next, this fast decay of the higher harmonics implies that communication is dominated by the $\ell = 0$ mode.

S6.3 Electrical signaling

The effect of the sender's signal is to add an additional current to the dynamics of the charge density field. With the signaling current, the dynamics become

$$\partial_t \lambda_\ell^m(t) = -\frac{1}{\tau_\ell}\lambda_\ell^m(t) + J_\ell^m(t). \tag{S6.15}$$

Writing these dynamics in frequency space and using the form of J in Eq. (S6.5) yields

$$\lambda_\ell^m(\omega) = \frac{\gamma_{\ell m}^I(R\hat{\mathbf{z}})\tau_\ell}{2\pi\sigma_I^2(1+i\omega\tau_l)} I(\omega). \tag{S6.16}$$

Here we have used $\gamma_{\ell m}^I(\mathbf{y})$ to denote the harmonic components of $\gamma^I(\mathbf{x},\mathbf{y})$, transforming the first coordinate. Having the response of λ_ℓ^m to the input signal, we can characterize the signal transmitted in our two communication schemes.

S6.3.1 Whole-cell sensing

In this case, our measurement is the total charge on the surface of the sphere. Note that we can write this as

$$M_{\text{cell}}(\omega) = \int ds(\mathbf{x})\, \lambda(\mathbf{x},\omega) = 2\pi^{1/2}R^2\lambda_0^0(\omega) = \frac{R^2\gamma_{00}^I(R\hat{\mathbf{z}})\tau_{\text{leak}}}{\pi^{1/2}\sigma_I^2(1+i\omega\tau_{\text{leak}})} I(\omega). \tag{S6.17}$$

Using Lemma S7.1 to approximate $\gamma_{00}^I(R\hat{\mathbf{z}})$, we obtain the response kernel of $M_{\text{cell}}(\omega)$ to $I(\omega)$:

$$\chi_{\text{cell}}^{MI}(\omega) \approx \frac{\tau_{\text{leak}}}{1+i\omega\tau_{\text{leak}}}. \tag{S6.18}$$

S6.3.2 Multiple sensors

The measurement of channel j in frequency space is

$$M_j(\omega) = \int ds(\mathbf{x})\, \gamma^M(\mathbf{x},\mathbf{y}_j)\lambda(\mathbf{x},\omega). \tag{S6.19}$$

We can decompose the response of M_j to I into contributions form different harmonics. To do this, note that the measurement can be seen as a convolution of λ and γ^M. The components of this convolution are computed in Lemma S7.2, giving the following decomposition:

$$M_j(\omega) = \sum_{\ell,m} R^2 \frac{2\pi^{1/2}}{(2\ell+1)^{1/2}}\gamma_{\ell 0}^M(R\hat{\mathbf{z}})\lambda_\ell^m Y_\ell^m(\mathbf{y}_j). \tag{S6.20}$$

Now, note that λ only has non-vanishing $m = 0$ harmonics, since the signal is rotationally symmetric around $\hat{\mathbf{z}}$. Using the response of λ_ℓ^0 to I and the harmonic representation of distances in Lemma S7.1, we get the following decomposition:

$$\chi_j^{MI}(\omega) \approx \frac{\sigma^2}{2R^2}\left(\frac{\tau_{\text{leak}}}{1+i\omega\tau_{\text{leak}}}\right) + \frac{\sigma^2}{2R^2}\sum_{\ell>0}\left(\frac{\tau_\ell}{1+i\omega\tau_\ell}\right)(2\ell+1)e^{-(\ell+1/2)^2(\sigma^2+\sigma_I^2)/2R^2}P_\ell(\cos(\theta_j)), \quad \text{(S6.21)}$$

where $\chi_j^{MI}(\omega)$ is the response kernel of $M_j(\omega)$ to $I(\omega)$.

For the parameters in the main text, we have $\tau_\ell^{-1} > \alpha/Rc \approx 10^7$ s-1. Thus, for $\omega \ll 10^7$ s-1, we have $\frac{\tau_\ell}{1+i\omega\tau_\ell} \approx \tau_\ell \approx \frac{Rc}{\alpha}\frac{2\ell+1}{\ell(\ell+1)}$, which gives us the approximation

$$\chi_j^{MI}(\omega) \approx \frac{\sigma^2}{2R^2}\left(\frac{\tau_{\text{leak}}}{1+i\omega\tau_{\text{leak}}}\right) + \frac{\sigma^2 c}{2\alpha R}\sum_{\ell>0}\frac{(2\ell+1)^2}{\ell(\ell+1)}e^{-(\ell+1/2)^2(\sigma^2+\sigma_I^2)/2R^2}P_\ell(\cos(\theta_j)). \quad \text{(S6.22)}$$

We can analyze the behavior of this response kernel for small angles. In this regime, the following approximation for the Legendre polynomials holds:

$$P_\ell(\cos(\theta)) = \sqrt{\frac{\theta}{\sin(\theta)}}J_0((\ell+1/2)\theta) + \mathcal{O}(\ell^{-1}), \quad \text{(S6.23)}$$

where J_0 is the Bessel function of the first kind. Despite this being a large ℓ expansion, taking the leading order yields a very good approximation for small θ, even for $\ell = 1$. Using this approximation and approximating the sum over ℓ with an integral, we obtain

$$\chi_j^{MI}(\omega) \approx \frac{\sigma^2}{2R^2}\left(\frac{\tau_{\text{leak}}}{1+i\omega\tau_{\text{leak}}}\right) + \frac{2\sigma^2 c}{\alpha R}\sqrt{\frac{\theta_j}{\sin(\theta_j)}}\int_0^\infty du\, e^{-u^2(\sigma^2+\sigma_I^2)/2R^2}J_0(u\theta_j). \quad \text{(S6.24)}$$

We can now use the identity (Eq. 6.631.1 in [2])

$$\int_0^\infty du\, e^{-au^2}J_0(bx) = \frac{1}{2}\left(\frac{\pi}{a}\right)^{1/2}e^{-b^2/8a}I_0\left(\frac{b^2}{8a}\right), \quad \text{(S6.25)}$$

where I_0 is the modified Bessel function of the first kind. Thus, our approximation now is

$$\chi_j^{MI}(\omega) \approx \frac{\sigma^2}{2R^2}\left(\frac{\tau_{\text{leak}}}{1+i\omega\tau_{\text{leak}}}\right) + \frac{\pi^{1/2}\sigma c}{\alpha}\left(\frac{2\sigma^2}{\sigma^2+\sigma_I^2}\right)^{1/2}\sqrt{\frac{\theta_j}{\sin(\theta_j)}}e^{-R^2\theta_j^2/4(\sigma^2+\sigma_I^2)}I_0\left(\frac{R^2\theta_j^2}{4(\sigma^2+\sigma_I^2)}\right). \quad \text{(S6.26)}$$

The function $e^{-x}I_0(x)$ behaves as $(2\pi x)^{-1/2}$ for large x. Therefore, for $\frac{\max\{\sigma,\sigma_I\}}{R} \ll \theta \ll 1$, the expression above scales as

$$\chi_j^{MI}(\omega) \approx \frac{\sigma^2}{2R^2}\left(\frac{\tau_{\text{leak}}}{1+i\omega\tau_{\text{leak}}}\right) + \frac{2\sigma c}{\alpha}\frac{\sigma}{R\theta_j} \quad \text{(S6.27)}$$

If $\omega\tau_{\text{leak}}$ is $\mathcal{O}(1)$, then for $\theta_j > \frac{Rc}{\alpha\tau_{\text{leak}}} \approx 10^{-4}$ the second term in the response function is dominated with respect to the first. Thus, the signal for most of the cell is driven by the zeroth harmonic:

$$\chi_j^{MI}(\omega) \approx \frac{\sigma^2}{2R^2}\left(\frac{\tau_{\text{leak}}}{1+i\omega\tau_{\text{leak}}}\right) = \frac{\sigma^2}{2R^2}\chi_{\text{cell}}^{MI}(\omega). \quad \text{(S6.28)}$$

S6.4 Johnson-Nyquist noise

We now calculate the fluctuations in our continuum model of charge dynamics. To do this, we use the fluctuation-dissipation theorem. The energy of the system is the energy stored in the membrane capacitor. We perturb this energy by coupling the charge density λ to a conjugate field h in the Hamiltonian:

$$\mathcal{H}[\lambda, h] = \int ds(\mathbf{x})\left[\frac{1}{2}\frac{\lambda^2(\mathbf{x})}{c} - h(\mathbf{x})\lambda(\mathbf{x})\right]. \quad \text{(S6.29)}$$

We do this so we can characterize the response of the charge to the perturbation, which can be linked to the noise spectrum via the fluctuation-dissipation theorem. The functional derivative of the perturbed Hamiltonian is

$$\frac{\delta \mathcal{H}[\lambda, h]}{\delta \lambda(\mathbf{x})} = \frac{\lambda(\mathbf{x})}{c} - h(\mathbf{x}). \tag{S6.30}$$

The perturbed dynamics are obtained by making the substitution

$$\frac{\lambda(\mathbf{x})}{c} \to \frac{\delta \mathcal{H}[\lambda, h]}{\delta \lambda(\mathbf{x})}. \tag{S6.31}$$

Finally, our perturbed dynamics (in the absence of a signal) are

$$\partial_t \lambda_\ell^m(t) = -\frac{1}{\tau_\ell}\lambda_\ell^m(t) - \frac{c}{\tau_\ell} h_\ell^m(t). \tag{S6.32}$$

To apply the fluctuation-dissipation theorem, we need to compute the response function of λ to h. We can do this by writing Eq. (S6.32) in Fourier space, which yields the response function

$$\chi_{\ell m}^{\lambda h}(\omega) = -\frac{c}{1 + i\omega\tau_\ell}. \tag{S6.33}$$

The fluctuation-dissipation theorem applied to the spherical setting is given in Lemma S7.3. Using our response function, the fluctuations in λ are

$$\tilde{S}_{\ell m}^{\lambda}(\omega) = (2\ell+1)^{1/2} \frac{c\tau_\ell}{\pi^{1/2} R^2 (1 + \omega^2 \tau_\ell^2)} \delta_{m0} k_B T. \tag{S6.34}$$

Explicitly, $\tilde{S}^\lambda(\mathbf{x}, \omega)$ is the Fourier transform of $\tilde{C}^\lambda(\mathbf{x}, t) \coloneqq C^\lambda(\mathbf{x}, R\hat{\mathbf{z}}, t)$, where

$$C^\lambda(\mathbf{x}, \mathbf{y}, t) \coloneqq \langle \lambda(\mathbf{x}, t_0)\lambda(\mathbf{y}, t_0 + t)\rangle. \tag{S6.35}$$

This expectation is taken with respect to the equilibrium fluctuations of the system. With the structure of the fluctuations of λ, we cna compute the fluctuations in our measurements.

S6.4.1 Whole-cell sensing

First, let us compute the temporal correlation function of the measurements of the cell

$$C_{\text{cell}}^M(t) = \langle M_{\text{cell}}(t_0) M_{\text{cell}}(t_0 + t)\rangle. \tag{S6.36}$$

Writing out the measurement of the cell explicitly,

$$C_{\text{cell}}^M(t) = \int ds(\mathbf{x})\, ds(\mathbf{y})\, \langle \lambda(\mathbf{x}, t_0)\lambda(\mathbf{y}, t_0 + t)\rangle = \int ds(\mathbf{x})\, ds(\mathbf{y})\, C^\lambda(\mathbf{x}, \mathbf{y}, t). \tag{S6.37}$$

Since the correlation function can only depend on $\mathbf{x} \cdot \mathbf{y}$, we can use rotational symmetry to reduce the expression above to a single integral:

$$C_{\text{cell}}^M(t) = 4\pi R^2 \int ds(\mathbf{x})\, C^\lambda(\mathbf{x}, R\hat{\mathbf{z}}, t) = 4\pi R^2 \int ds(\mathbf{x})\, \tilde{C}^\lambda(\mathbf{x}, t) = 8\pi^{3/2} R^4 \tilde{C}_{00}^\lambda(t). \tag{S6.38}$$

Taking the Fourier transform and using Eq. (S6.34), the fluctuation spectrum of the cell measurement is

$$S_{\text{cell}}^M(\omega) = \frac{8\pi R^2 c\tau_{\text{leak}}}{(1 + \omega^2 \tau_{\text{leak}}^2)} k_B T. \tag{S6.39}$$

S6.4.2 Multiple sensors

First, we compute the fluctuations in the measurement made by a single sensor. Since there is no signal, we can set $\mathbf{y}_j = R\hat{\mathbf{z}}$ due to rotational symmetry. The correlations in the channel measurement are

$$\begin{aligned} C^M_{\text{channel}}(t) &= \int ds(\mathbf{x})\, ds(\mathbf{y})\, \gamma^M(\mathbf{x}, R\hat{\mathbf{z}})\gamma^M(\mathbf{y}, R\hat{\mathbf{z}})\langle\lambda(\mathbf{x}, t_0)\lambda(\mathbf{y}, t_0 + t)\rangle \\ &= \int ds(\mathbf{x})\, ds(\mathbf{y})\, \gamma^M(\mathbf{x}, R\hat{\mathbf{z}})\gamma^M(\mathbf{y}, R\hat{\mathbf{z}})C^\lambda(\mathbf{x}, \mathbf{y}, t) \end{aligned} \tag{S6.40}$$

We can write the $\mathbf{y}$ integral as a convolution and use Lemma S7.2 to represent its components. This gives us a decomposition of the correlation function:

$$C^M_{\text{channel}}(t) = 2\pi^{1/2}R^4 \sum_{\ell,m} \frac{(-1)^m}{(2\ell+1)^{1/2}} \gamma^M_{\ell m}(R\hat{\mathbf{z}})\gamma^M_{\ell,-m}(R\hat{\mathbf{z}})\tilde{C}^\lambda_{\ell 0}(t). \tag{S6.41}$$

Taking the Fourier transform and using Lemma S7.1 to write our distance factors, we get a decomposition of the channel noise spectrum:

$$S^M_{\text{channel}}(\omega) \approx 2\pi^{3/2}\sigma^4 \left[\tilde{S}^\lambda_{00}(\omega) + \sum_{\ell>0} (2\ell+1)^{1/2} e^{-(\ell+1/2)^2\sigma^2/R^2} \tilde{S}^\lambda_{\ell 0}(\omega)\right]. \tag{S6.42}$$

With our expression for the λ spectrum, this decomposition becomes

$$S^M_{\text{channel}}(\omega) \approx \frac{2\pi c\sigma^4}{R^2} k_B T \left[\frac{\tau_{\text{leak}}}{1+\omega^2\tau^2_{\text{leak}}} + \sum_{\ell>0} \frac{(2\ell+1)\tau_\ell}{1+\omega^2\tau^2_\ell} e^{-(\ell+1/2)^2\sigma^2/R^2}\right]. \tag{S6.43}$$

The decomposition above is useful to understand the sources of noise in the channel's measurements. The first term corresponds to the noise from the measurement $(\sigma^2/2R^2)M_{\text{cell}}$, so we can interpret it as the collective noise coming from the fluctuations in the zeroth mode. The rest of the decomposition corresponds to the local Johnson-Nyquist fluctuations. To see this, suppose that ω is small compared to the first few τ_ℓ. Then we can approximate the sum as

$$\sum_{\ell>0} \frac{(2\ell+1)\tau_\ell}{1+\omega^2\tau^2_\ell} e^{-(\ell+1/2)^2\sigma^2/R^2} \approx \frac{Rc}{\alpha} \sum_{\ell>0} (2\ell+1) \frac{2\ell+1}{\ell(\ell+1)} e^{-(\ell+1/2)^2\sigma^2/R^2}. \tag{S6.44}$$

We can further approximate this by evaluating the sum as an integral:

$$\sum_{\ell>0} \frac{(2\ell+1)\tau_\ell}{1+\omega^2\tau^2_\ell} e^{-(\ell+1/2)^2\sigma^2/R^2} \approx \frac{Rc}{\alpha} \int_1^\infty du\, \frac{(2u+1)^2}{u(u+1)} e^{-(u+1/2)^2\sigma^2/R^2} \approx \frac{R^2 c}{\alpha\sigma} \int_0^\infty dv \frac{(2v)^2}{v^2} e^{-v^2}. \tag{S6.45}$$

Evaluating this integral and substituting into our expression of fluctuations, we get

$$S^M_{\text{channel}}(\omega) \approx \frac{2\pi c\sigma^4}{R^2} \left(\frac{\tau_{\text{leak}}}{1+\omega^2\tau^2_{\text{leak}}}\right) k_B T + \frac{4\pi^{3/2}c^2\sigma^3}{\alpha} k_B T. \tag{S6.46}$$

The second term is precisely the single-channel Johnson-Nyquist from Equation (S4.9). Therefore, a single ion channel experiences noise from the slow fluctuations in the zeroth harmonic and from the local fluctuations we characterized in Section S3.

To conclude our characterization of noise in the multiple-channel measurement, we show that fluctuations in different channels are only correlated through the fluctuations in the total charge. For this, we compute the correlation function between the measurements of channels j and k:

$$C^M_{jk}(t) = \langle M_j(t_0) M_k(t_0 + t)\rangle. \tag{S6.47}$$

Due to rotational symmetry, we can perform a rotation such that $\mathbf{y}_j$ gets sent to $R\hat{\mathbf{z}}$. Under this rotation, $\mathbf{y}_k$ gets sent to a new position, which we denote $\mathbf{y}_{jk}$. Following the same procedure as for our previous noise spectra, we find the following decomposition of the cross spectrum:

$$S^M_{jk}(\omega) = 2\pi^{1/2}R^4 \sum_\ell (2\ell+1)^{-1/2}\gamma^M_{\ell 0}(R\hat{\mathbf{z}})\gamma^M_{\ell 0}(\mathbf{y}_{jk})\tilde{S}^\lambda_{\ell 0}(\omega). \tag{S6.48}$$

Using Lemma S7.4, we can obtain the scaling of this spectrum in θ_{jk}, the angular distance between $\mathbf{y}_j$ and $\mathbf{y}_k$:

$$S^M_{jk}(\omega) \approx \frac{2\pi c\sigma^4}{R^2}\left(\frac{\tau_{\text{leak}}}{1+\omega^2\tau^2_{\text{leak}}}\right) + \frac{2\pi c\sigma^4}{R^2}k_BT\sum_{\ell>0}\frac{\tau_\ell}{1+\omega^2\tau^2_\ell}(2\ell+1)e^{-(\ell+1/2)^2\sigma^2/R^2}J_0((\ell+1/2)\theta_{jk}). \tag{S6.49}$$

Note that for small ω this sum is very similar to the one in Equation (S6.45), with an added Bessel function. Therefore, we can perform the same integral approximation and use Eq. (S6.25) to obtain

$$\begin{aligned}\sum_{\ell>0}\frac{(2\ell+1)\tau_\ell}{1+\omega^2\tau^2_\ell}e^{-(\ell+1/2)^2\sigma^2/R^2}J_0((\ell+1/2)\theta_{jk}) &\approx \frac{4R^2c}{\alpha\sigma}\int_0^\infty dv\, e^{-v^2}J_0\left(\frac{R}{\sigma}v\theta_{jk}\right)\\ &= \frac{2\pi^{1/2}R^2c}{\alpha\sigma}e^{-\theta^2_{jk}R^2/8\sigma^2}I_0\left(\frac{\theta^2_{jk}R^2}{8\sigma^2}\right).\end{aligned} \tag{S6.50}$$

Thus, our cross-spectrum has the following behavior:

$$S^M_{jk}(\omega) \approx \frac{2\pi c\sigma^4}{R^2}\left(\frac{\tau_{\text{leak}}}{1+\omega^2\tau^2_{\text{leak}}}\right)k_BT + \frac{4\pi^{3/2}c^2\sigma^3}{\alpha}k_BT\cdot e^{-\theta^2_{jk}R^2/8\sigma^2}I_0\left(\frac{\theta^2_{jk}R^2}{8\sigma^2}\right). \tag{S6.51}$$

The function $e^{-x}I_0(x)$ behaves as $(2\pi x)^{-1/2}$ for large x, so the local Johnson-Nyquist contribution is suppressed if the channels that are separated more than a few σ. Therefore, for separated channels only the first term survives. This cross-spectrum is consistent with the model in the main text, in which every channel experiences a common source of noise from the zeroth harmonic and independent local noise. Since we know shot noise dominates the local component, we can assume the local measurement noise is simply shot noise.

S7 Technical Lemmas

S7.1 Harmonic representation of distances

Lemma: $\gamma^{I,M}_{\ell m}(R\hat{\mathbf{z}})$ formula

The gaussian distance factor $\gamma^I(\mathbf{x}, R\hat{\mathbf{z}})$ has harmonic components

$$\gamma^I_{\ell m}(R\hat{\mathbf{z}}) = \delta_{m0}\pi^{1/2}(2\ell+1)^{1/2}\left[\int_{-1}^1 du\, e^{-(1-u)R^2/\sigma_I^2}P_\ell(u)\right], \tag{S7.1}$$

where P_ℓ is the ℓ-th Legendre polynomial. Additionally, in the $\sigma_I/R \ll 1$ regime, these can be approximated by

$$\gamma^I_{\ell 0}(R\hat{\mathbf{z}}) \approx \begin{cases}\pi^{1/2}(\sigma_I^2/R^2) & \text{if } \ell = 0,\\ \pi^{1/2}(\sigma_I^2/R^2)(2\ell+1)^{1/2}e^{-(\ell+1/2)^2\sigma_I^2/2R^2} & \text{if } \ell > 0.\end{cases} \tag{S7.2}$$

Analogous results hold for γ^M, replacing σ_I with σ.

The harmonic component $\gamma^I_{\ell m}(R\hat{\mathbf{z}})$ is given by

$$\gamma^I_{\ell m}(R\hat{\mathbf{z}}) = \frac{1}{R^2}\int ds(\mathbf{x})\,\gamma^I(\mathbf{x}, R\hat{\mathbf{z}})Y^{m*}_\ell(\mathbf{x}). \tag{S7.3}$$

Since $\gamma^I(\mathbf{x}, R\hat{\mathbf{z}})$ has rotational symmetry about the north pole, we have that all the $m \neq 0$ components vanish. For the $m = 0$ components, we can write the integral in terms of Legendre polynomials using spherical coordinates:

$$\begin{aligned}\gamma^I_{\ell m}(R\hat{\mathbf{z}}) &= \frac{e^{-R^2/\sigma_I^2}}{R^2}\delta_{m0}\int ds(\mathbf{x})\, e^{R\mathbf{x}\cdot\hat{\mathbf{z}}/\sigma_I^2} Y_\ell^{0*}(\mathbf{x}) \\ &= \frac{e^{-R^2/\sigma_I^2}}{R^2}\delta_{m0}(2\ell+1)^{1/2}\int_0^\pi d\theta\, \sin(\theta) e^{R^2\cos(\theta)/\sigma_I^2} P_\ell(\cos(\theta)).\end{aligned} \tag{S7.4}$$

Substituting $u = \cos(\theta)$, we obtain the desired result:

$$\gamma^I_{\ell m}(R\hat{\mathbf{z}}) = \delta_{m0}\pi^{1/2}(2\ell+1)^{1/2}\left[\int_{-1}^1 du\, e^{-(1-u)R^2/\sigma_I^2} P_\ell(u)\right]. \tag{S7.5}$$

For $\ell = 0$, we have $P_0(u) = 1$. Since we are working in the $\sigma_I/R \ll 1$ regime, we can make the following approximation:

$$\int_{-1}^1 du\, e^{-(1-u)R^2/\sigma_I^2} \approx \int_0^\infty dv\, e^{-vR^2/\sigma_I^2} = \frac{\sigma_I^2}{R^2}. \tag{S7.6}$$

Therefore, our zeroth harmonic can be approximated by

$$\gamma^I_{\ell 0}(R\hat{\mathbf{z}}) = \pi^{1/2}\frac{\sigma_I^2}{R^2}. \tag{S7.7}$$

For higher harmonics, we can approximate $\gamma^I_{\ell 0}$ by using the asymptotic behavior of the Legendre polynomials:

$$P_\ell(\cos(\theta)) = \sqrt{\frac{\theta}{\sin(\theta)}} J_0((\ell+1/2)\theta) + \mathcal{O}(\ell^{-1}), \tag{S7.8}$$

where J_0 is the Bessel function of the first kind. Despite this being a large ℓ expansion, taking the leading order yields a very good approximation for small θ, even for $\ell = 1$. Thus,

$$\int_{-1}^1 du\, e^{-(1-u)R^2/\sigma_I^2} P_\ell(u) \approx \int_0^\pi d\theta\, \sqrt{\theta\sin(\theta)}\, e^{-(1-\cos(\theta))R^2/\sigma_I^2} J_0((\ell+1/2)\theta). \tag{S7.9}$$

Again, since $\sigma_I/R \ll 1$, most of the mass concentrates near $\theta = 0$, meaning that we can take the upper limit to ∞ and expand the functions that do not depend on ℓ to leading order:

$$\int_{-1}^1 du\, e^{-(1-u)R^2/\sigma_I^2} P_\ell(u) \approx \int_0^\infty dy\, y e^{-y^2R^2/2\sigma_I^2} J_0((\ell+1/2)y) = \frac{\sigma_I^2}{R^2} e^{-(\ell+1/2)^2\sigma_I^2/2R^2}. \tag{S7.10}$$

This yields our approximation for $\ell > 0$:

$$\gamma^I_{\ell 0}(R\hat{\mathbf{z}}) \approx \pi^{1/2}\frac{\sigma_I^2}{R^2}(2\ell+1)^{1/2} e^{-(\ell+1/2)^2\sigma_I^2/2R^2}. \tag{S7.11}$$

S7.2 Spherical Convolution

Lemma: Spherical Convolution

Let f be defined by

$$f(\mathbf{x}) = \int ds(\mathbf{x}')\, h(\mathbf{x}') g(\mathbf{x} \cdot \mathbf{x}'). \tag{S7.12}$$

Then the harmonic components of f are

$$f_\ell^m = R^2 \sqrt{\frac{4\pi}{2\ell+1}} h_\ell^m \tilde{g}_\ell^0, \tag{S7.13}$$

where $\tilde{g}(\mathbf{x}) \coloneqq g(\mathbf{x} \cdot R\hat{\mathbf{z}})$.

To simplify the proof, we will use Dirac notation. For a given function $f(\mathbf{x})$, define the associated state $|f\rangle$ by

$$|f\rangle \equiv \sum_{\ell,m} f_\ell^m \, |\ell, m\rangle \, . \tag{S7.14}$$

Additionally, we have that the space representation of this state is given by $f(\mathbf{x}) = \langle \mathbf{x} | f \rangle$, where the states $|\mathbf{x}\rangle$ satisfy the completeness relation

$$\frac{1}{R^2} \int ds(\mathbf{x}) \; |\mathbf{x}\rangle \, \langle \mathbf{x}| = \mathbb{1}. \tag{S7.15}$$

We can, therefore, write the integral expression for f as

$$\langle \mathbf{x} | f \rangle = \int ds(\mathbf{x}') \; \langle \mathbf{x}' | h \rangle \, g(\mathbf{x} \cdot \mathbf{x}'). \tag{S7.16}$$

By making a change of variables $\mathbf{x}' \to \mathbf{A}(\mathbf{x})\mathbf{x}'$ using a rotation matrix $\mathbf{A}(\mathbf{x})$ that satisfies $\mathbf{A}^{-1}(\mathbf{x})\mathbf{x} = R\hat{\mathbf{z}}$, we can write the integral as

$$\begin{aligned} \langle \mathbf{x} | f \rangle &= \int ds(\mathbf{x}') \; \langle \mathbf{A}(\mathbf{x})\mathbf{x}' | h \rangle \, g(R\mathbf{x}' \cdot \hat{\mathbf{z}}) \\ &= \int ds(\mathbf{x}') \; \langle \mathbf{A}(\mathbf{x})\mathbf{x}' | h \rangle \, \tilde{g}(\mathbf{x}') \\ &= \int ds(\mathbf{x}') \; \langle \mathbf{A}(\mathbf{x})\mathbf{x}' | h \rangle \, \langle \mathbf{x}' | \tilde{g} \rangle \, . \end{aligned} \tag{S7.17}$$

Since h is a real function, it must satisfy $\langle \mathbf{A}(\mathbf{x})\mathbf{x}' | h \rangle = \langle h | \mathbf{A}(\mathbf{x})\mathbf{x}' \rangle$. Additionally, we have that in state space, rotations act on kets as $|\mathbf{A}(\mathbf{x})\mathbf{x}'\rangle = \mathcal{A}(\mathbf{x}) \, |\mathbf{x}'\rangle$, where $\mathcal{A}(\mathbf{x})$ is the associated rotation operator. Using this and the completeness relation, we obtain

$$\langle \mathbf{x} | f \rangle = R^2 \, \langle h | \mathcal{A}(\mathbf{x}) | \tilde{g} \rangle \, . \tag{S7.18}$$

In order to get the harmonic components of f, we integrate the above expression, which yields

$$\begin{aligned} \langle \ell, m | f \rangle &= \int ds(\mathbf{x}) \; \langle \ell, m | \mathbf{x} \rangle \, \langle h | \mathcal{A}(\mathbf{x}) | \tilde{g} \rangle \\ &= \sum_{\ell', m'} \sum_{\ell'', m''} \langle h | \ell', m' \rangle \, \langle \ell'', m'' | \tilde{g} \rangle \int ds(\mathbf{x}) \; \langle \ell, m | \mathbf{x} \rangle \, \langle \ell', m' | \mathcal{A}(\mathbf{x}) | \ell'', m'' \rangle \, . \end{aligned} \tag{S7.19}$$

Now, note that, since $\tilde{g}$ has rotational symmetry about the north pole, $\langle \ell'', m''|\tilde{g}\rangle = \langle \ell'', 0|\tilde{g}\rangle\, \delta_{m''0}$. This implies that

$$\langle \ell, m|f\rangle = \sum_{\ell',m'} \sum_{\ell''} \langle h|\ell', m'\rangle \langle \ell'', 0|\tilde{g}\rangle \int ds(\mathbf{x})\ \langle \ell, m|\mathbf{x}\rangle \langle \ell', m'|\mathcal{A}(\mathbf{x})|\ell'', 0\rangle\,. \tag{S7.20}$$

Since the subspace associated to the ℓ harmonics is invariant under rotations, we know that $\langle \ell', m'|\mathcal{A}(\mathbf{x})|\ell'', 0\rangle \propto \delta_{\ell'\ell''}$. we, then, have that

$$\langle \ell, m|f\rangle = \sum_{\ell',m'} \langle h|\ell', m'\rangle \langle \ell', 0|g\rangle \int ds(\mathbf{x})\ \langle \ell, m|\mathbf{x}\rangle \langle \ell', m'|\mathcal{A}(\mathbf{x})|\ell', 0\rangle\,. \tag{S7.21}$$

Let $|R\hat{\mathbf{z}}\rangle$ denote the position state corresponding to the north pole. We have that, for a given ℓ, the only spherical harmonic that has a non-zero component at the north pole corresponds to $m = 0$. Specifically, we have that

$$\langle R\hat{\mathbf{z}}|\ell, m\rangle = \sqrt{\frac{2\ell+1}{4\pi}}\delta_{m0} = \sqrt{\frac{2\ell+1}{4\pi}}\,\langle \ell, 0|\ell, m\rangle\,. \tag{S7.22}$$

Additionally, note that since $\mathbf{x} = \mathbf{A}(\mathbf{x})(R\hat{\mathbf{z}})$, we can write

$$\langle \mathbf{x}|\ell, m\rangle = \langle R\hat{\mathbf{z}}|\mathcal{A}(\mathbf{x})^\dagger|\ell, m\rangle\,. \tag{S7.23}$$

Since the ℓ subspace is invariant under rotations, we have that $\mathcal{A}(\mathbf{x})^\dagger\,|\ell, m\rangle$ is a linear combination of ℓ harmonics. Given the previous result, this implies that

$$\langle \mathbf{x}|\ell, m\rangle = \sqrt{\frac{2\ell+1}{4\pi}}\,\langle \ell, 0|\mathcal{A}(\mathbf{x})^\dagger|\ell, m\rangle\,. \tag{S7.24}$$

Taking the conjugate of this expression, we obtain

$$\langle \ell, m|\mathbf{x}\rangle = \sqrt{\frac{2\ell+1}{4\pi}}\,\langle \ell, m|\mathcal{A}(\mathbf{x})|\ell, 0\rangle\,. \tag{S7.25}$$

Plugging this into the expression for f_ℓ^m:

$$\langle \ell, m|f\rangle = \sum_{\ell',m'} \sqrt{\frac{4\pi}{2\ell'+1}}\,\langle h|\ell', m'\rangle \langle \ell', 0|\tilde{g}\rangle \int ds(\mathbf{x})\ \langle \ell, m|\mathbf{x}\rangle \langle \ell', m'|\mathbf{x}\rangle\,. \tag{S7.26}$$

The spherical harmonics satisfy

$$\langle \ell, m|\mathbf{x}\rangle = (-1)^m\,\langle \mathbf{x}|\ell, -m\rangle\,. \tag{S7.27}$$

This yields the following expression:

$$\begin{aligned}
\langle \ell, m|f\rangle &= \sum_{\ell',m'} (-1)^{m'} \sqrt{\frac{4\pi}{2\ell'+1}}\,\langle h|\ell', m'\rangle \langle \ell', 0|\tilde{g}\rangle \int ds(\mathbf{x})\ \langle \ell, m|\theta, \phi\rangle \langle \theta, \phi|\ell', -m'\rangle \\
&= R^2 \sum_{\ell',m'} (-1)^{m'} \sqrt{\frac{4\pi}{2\ell'+1}}\,\langle h|\ell', m'\rangle \langle \ell', 0|\tilde{g}\rangle \langle \ell, m|\ell', -m'\rangle \\
&= R^2 (-1)^m \sqrt{\frac{4\pi}{2\ell+1}}\,\langle h|\ell, -m\rangle \langle \ell, 0|\tilde{g}\rangle \\
&= R^2 \sqrt{\frac{4\pi}{2\ell+1}}\,\langle \ell, m|h\rangle \langle \ell, 0|\tilde{g}\rangle\,,
\end{aligned} \tag{S7.28}$$

where the last equality follows from the same conjugation property used before. Switching back to the original notation, we have that

$$f_\ell^m = R^2 \sqrt{\frac{4\pi}{2\ell+1}} h_\ell^m \tilde{g}_\ell^0. \tag{S7.29}$$

Note that f corresponds to the isotropic convolution of h and g, as defined in [3].

S7.3 Spherical fluctuation-dissipation theorem

Lemma: Fluctuation-dissipation theorem on a sphere

Let a system have a hamiltonian given by the following functional:

$$H[\lambda] = H_0[\lambda] - \int ds(\mathbf{x})\, \lambda(\mathbf{x}) h(\mathbf{x}), \tag{S7.30}$$

where h is the conjugate field. Additionally, suppose that the harmonic components have the following impulse-response behavior:

$$\langle \lambda_\ell^m(t) \rangle = \langle \lambda_\ell^m \rangle_0 + \int_{-\infty}^{t} d\tau\, h_\ell^m(\tau) \chi_{\ell m}^{\lambda h}(t-\tau), \tag{S7.31}$$

where $\langle \cdot \rangle_0$ denotes the expectation in the absence of a field. Then the spectrum of the equilibrium fluctuations in the λ field is given by

$$\tilde{S}_{\ell m}^{\lambda}(\omega) = -\sqrt{\frac{2\ell+1}{4\pi}} \frac{2k_B T}{\omega R^2} \mathrm{Im}\left[\chi_{\ell 0}^{\lambda h}(\omega)\right] \delta_{m0}, \tag{S7.32}$$

where $\tilde{S}^{\lambda}(\mathbf{x}, \omega)$ is the Fourier transform of $\tilde{C}^{\lambda}(\mathbf{x}, t) \coloneqq C^{\lambda}(\mathbf{x}, R\hat{\mathbf{z}}, t)$, and C^{λ} is the equilibrium correlation function.

Consider the case where the h field is active for all $t \leq 0$ and is turned off at $t = 0$:

$$h(\mathbf{x}, t) = h_0(\mathbf{x}) \Theta(-t), \tag{S7.33}$$

where Θ is the Heaviside step function. For $t > 0$, we can write the expectation of the λ field as

$$\langle \lambda(\mathbf{x}, t) \rangle = \int \mathcal{D}\lambda'\, \mathcal{D}\lambda\, \lambda'(\mathbf{x}) \mathcal{P}(\lambda', t | \lambda, 0) \mathcal{W}(\lambda, 0), \tag{S7.34}$$

where $\mathcal{P}$ is the transition kernel and $\mathcal{W}$ is given by the Boltzmann distribution at $t = 0$:

$$\mathcal{W}(\lambda, 0) = \frac{e^{-\beta H[\lambda]}}{\int \mathcal{D}\lambda'\, e^{-\beta H[\lambda']}}. \tag{S7.35}$$

For a weak field, we have that

$$\mathcal{W}(\lambda, 0) \approx \mathcal{W}_0(\lambda, 0) \left[1 + \beta \int ds(\mathbf{x})\, (\lambda(\mathbf{x}) - \langle \lambda(\mathbf{x}) \rangle_0) h_0(\mathbf{x})\right]. \tag{S7.36}$$

Returning to the expression for $\langle \lambda \rangle$,

$$\begin{aligned} \langle \lambda(\mathbf{x}, t) \rangle &= \int \mathcal{D}\lambda'\, \mathcal{D}\lambda\, \lambda'(\mathbf{x}) \mathcal{P}(\lambda', t | \lambda, 0) \mathcal{W}_0(\lambda, 0) \left[1 + \beta \int ds(\mathbf{x}')\, (\lambda(\mathbf{x}') - \langle \lambda(\mathbf{x}') \rangle_0) h_0(\mathbf{x}')\right] \\ &= \langle \lambda(\mathbf{x}) \rangle_0 + \beta \int ds(\mathbf{x}')\, h_0(\mathbf{x}') [\langle \lambda(\mathbf{x}', 0) \lambda(\mathbf{x}, t) \rangle_0 - \langle \lambda(\mathbf{x}) \rangle_0 \langle \lambda(\mathbf{x}') \rangle_0]. \end{aligned} \tag{S7.37}$$

We can define the correlation function in equilibrium as

$$C^{\lambda}(\mathbf{x}, \mathbf{x}', t, t') = \langle \lambda(\mathbf{x}, t) \lambda(\mathbf{x}', t') \rangle_0 - \langle \lambda(\mathbf{x}) \rangle_0 \langle \lambda(\mathbf{x}') \rangle_0. \tag{S7.38}$$

Note that this correlation function should be invariant under rotations and under time translations, meaning that it can be fully specified by $\mathbf{x} \cdot \mathbf{x}'$, and by $|t - t'|$. We, therefore, define

$$C^{\lambda}(\mathbf{x} \cdot \mathbf{x}', |t - t'|) = \langle \lambda(\mathbf{x}, t) \lambda(\mathbf{x}', t') \rangle_0 - \langle \lambda(\mathbf{x}) \rangle_0 \langle \lambda(\mathbf{x}') \rangle_0. \tag{S7.39}$$

Plugging this into our previous expression yields

$$\langle \lambda(\mathbf{x},t)\rangle = \langle \lambda(\mathbf{x})\rangle_0 + \beta \int ds(\mathbf{x}')\, h_0(\mathbf{x}') C^{\lambda}(\mathbf{x}\cdot\mathbf{x}', t). \tag{S7.40}$$

Using Lemma (S7.2), we can write the integral as

$$\int ds(\mathbf{x}')\, h_0(\mathbf{x}') C^{\lambda}(\mathbf{x}\cdot\mathbf{x}', t) = R^2 \sum_{\ell,m} \sqrt{\frac{4\pi}{2\ell+1}} (h_0)_\ell^m \tilde{C}^{\lambda}_{\ell 0}(t) Y_\ell^m(\mathbf{x}), \tag{S7.41}$$

where $\tilde{C}^{\lambda}(\mathbf{x},t) = C^{\lambda}(R\mathbf{x}\cdot\hat{\mathbf{z}}, t)$. We, therefore, obtain the following expression for the harmonic components of λ:

$$\langle \lambda_\ell^m(t)\rangle = \langle \lambda_\ell^m\rangle_0 + \beta R^2 \sqrt{\frac{4\pi}{2\ell+1}} (h_0)_\ell^m \tilde{C}^{\lambda}_{\ell 0}(t). \tag{S7.42}$$

Comparison with Equation (S7.31) yields

$$\int_{-\infty}^{0} d\tau\, (h_0)_\ell^m \chi^{\lambda h}_{\ell m}(t-\tau) = \beta R^2 \sqrt{\frac{4\pi}{2\ell+1}} (h_0)_\ell^m \tilde{C}^{\lambda}_{\ell 0}(t). \tag{S7.43}$$

For this to be valid for any conjugate field, we must have that

$$\int_{-\infty}^{0} d\tau\, \chi^{\lambda h}_{\ell m}(t-\tau) = \beta R^2 \sqrt{\frac{4\pi}{2\ell+1}} \tilde{C}^{\lambda}_{\ell 0}(t). \tag{S7.44}$$

Since the correlation function is even in t and real, the power spectrum must satisfy

$$\tilde{S}^{\lambda}_{\ell 0}(\omega) = 2\mathrm{Re}\left[\int_0^\infty dt\, e^{-i\omega t} \tilde{C}^{\lambda}_{\ell 0}(t)\right]. \tag{S7.45}$$

Using the previous expression for the correlation function, we have that

$$\begin{aligned}
\tilde{S}^{\lambda}_{\ell 0}(\omega) &= \sqrt{\frac{2\ell+1}{4\pi}} \frac{2k_BT}{R^2} \mathrm{Re}\left[\int_0^\infty dt\, e^{-i\omega t} \int_0^\infty d\tau\, \chi^{\lambda h}_{\ell m}(t+\tau)\right] \\
&= \sqrt{\frac{2\ell+1}{4\pi}} \frac{2k_BT}{R^2} \mathrm{Re}\left[\int_0^\infty dt\, e^{-i\omega t} \int_0^\infty d\tau\, \chi^{\lambda h}_{\ell m}(\tau)\Theta(\tau-t)\right] \\
&= \sqrt{\frac{2\ell+1}{4\pi}} \frac{2k_BT}{R^2} \mathrm{Re}\left[\frac{i}{\omega}\int_0^\infty d\tau\, \chi^{\lambda h}_{\ell m}(\tau)\left(e^{-i\omega\tau}-1\right)\right] \\
&= \sqrt{\frac{2\ell+1}{4\pi}} \frac{2k_BT}{\omega R^2} \mathrm{Re}\left[i\left(\chi^{\lambda h}_{\ell m}(\omega) - \chi^{\lambda h}_{\ell m}(0)\right)\right].
\end{aligned} \tag{S7.46}$$

Since $\chi^{\lambda h}_{\ell m}(0)$ is real, we have that

$$\tilde{S}^{\lambda}_{\ell 0}(\omega) = -\sqrt{\frac{2\ell+1}{4\pi}} \frac{2k_BT}{\omega R^2} \mathrm{Im}\left[\chi^{\lambda h}_{\ell m}(\omega)\right]. \tag{S7.47}$$

Since this is valid for any m, we evaluate it specifically for $m = 0$.

S7.4 Harmonic representation of distances for arbitrary orientations

Lemma: $\gamma^{I,M}_{\ell 0}(\mathbf{y})$ formula

The gaussian distance factor $\gamma^I(\mathbf{x},\mathbf{y})$ has $m=0$ harmonic components

$$\gamma^I_{\ell 0}(\mathbf{y}) = \gamma^I_{\ell 0}(R\hat{\mathbf{z}})P_\ell(\cos(\theta)), \tag{S7.48}$$

where P_ℓ is the ℓ-th Legendre polynomial and θ is the polar angle of $\mathbf{y}$. For $\sigma_I/R \ll 1$ and $\theta \ll 1$, these can be approximated by

$$\gamma^I_{\ell 0}(\mathbf{y}) \approx \begin{cases} \pi^{1/2}(\sigma_I^2/R^2) & \text{if } \ell = 0, \\ \pi^{1/2}(\sigma_I^2/R^2)(2\ell+1)^{1/2} e^{-(\ell+1/2)^2\sigma_I^2/2R^2} J_0((\ell+1/2)\theta) & \text{if } \ell > 0. \end{cases} \tag{S7.49}$$

Analogous results hold for γ^M, replacing σ_I with σ.

Since $\gamma^I(\mathbf{x},\mathbf{y})$ only depends on $\mathbf{x}\cdot\mathbf{y}$, we can write

$$\gamma^I(\mathbf{x},\mathbf{y}) =: g\left(\frac{\mathbf{x}}{R}\cdot\frac{\mathbf{y}}{R}\right). \tag{S7.50}$$

That is, g is a function of $\cos(\theta)$, where θ is the angular distance between $\mathbf{x}$ and $\mathbf{y}$. Since the Legendre polynomials are complete on $[-1,1]$, g can be expanded as

$$g\left(\frac{\mathbf{x}}{R}\cdot\frac{\mathbf{y}}{R}\right) = \sum_\ell a_\ell P_\ell\left(\frac{\mathbf{x}}{R}\cdot\frac{\mathbf{y}}{R}\right). \tag{S7.51}$$

The Legendre polynomials satisfy the following identity:

$$P_\ell\left(\frac{\mathbf{x}}{R}\cdot\frac{\mathbf{y}}{R}\right) = \frac{4\pi}{2\ell+1}\sum_{m=-\ell}^{\ell} Y_\ell^m(\mathbf{x})Y_\ell^{m*}(\mathbf{y}). \tag{S7.52}$$

Therefore, $\gamma^I(\mathbf{x},\mathbf{y})$ admits the decomposition

$$\gamma^I(\mathbf{x},\mathbf{y}) = \sum_{\ell,m}\frac{4\pi}{2\ell+1}a_\ell Y_\ell^m(\mathbf{x})Y_\ell^{m*}(\mathbf{y}). \tag{S7.53}$$

Taking this expression as a function of $\mathbf{x}$, its harmonic components are

$$\gamma^I_{\ell m}(\mathbf{y}) = \frac{4\pi}{2\ell+1}a_\ell Y_\ell^{m*}(\mathbf{y}). \tag{S7.54}$$

To obtain the coefficients a_ℓ, we can evaluate our expression at $\mathbf{y} = R\hat{\mathbf{z}}$:

$$\gamma^I_{\ell m}(R\hat{\mathbf{z}}) = \frac{4\pi}{2\ell+1}a_\ell Y_\ell^{m*}(R\hat{\mathbf{z}}) = \frac{2\pi^{1/2}}{(2\ell+1)^{1/2}}a_\ell\delta_{m0} \implies a_\ell = \frac{(2\ell+1)^{1/2}}{2\pi^{1/2}}\gamma^I_{\ell 0}(R\hat{\mathbf{z}}). \tag{S7.55}$$

Using this expression in our harmonic components for γ^I and evaluating at $m=0$, we obtain

$$\gamma^I_{\ell 0}(\mathbf{y}) = \frac{2\pi^{1/2}}{(2\ell+1)^{1/2}}\gamma^I_{\ell 0}(R\hat{\mathbf{z}})Y_\ell^{0*}(\mathbf{y}) = \gamma^I_{\ell 0}(R\hat{\mathbf{z}})P_\ell(\cos(\theta)). \tag{S7.56}$$

Our approximation is obtained by approximating $\gamma^I_{\ell 0}(R\hat{\mathbf{z}})$ using Lemma S7.1 and approximating the Legendre polynomials by

$$P_\ell(\cos(\theta)) \approx \sqrt{\frac{\theta}{\sin(\theta)}}J_0((\ell+1/2)\theta). \tag{S7.57}$$

For small angles, we have

$$\sqrt{\frac{\theta}{\sin(\theta)}} = 1 + \mathcal{O}(\theta^2), \tag{S7.58}$$

so we can use the approximation $P_\ell(\cos(\theta)) \approx J_0((\ell + 1/2)\theta)$.

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